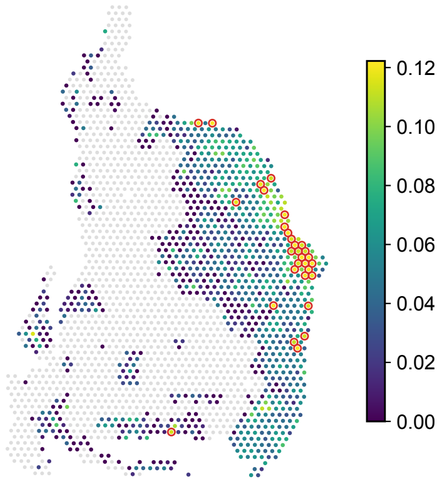
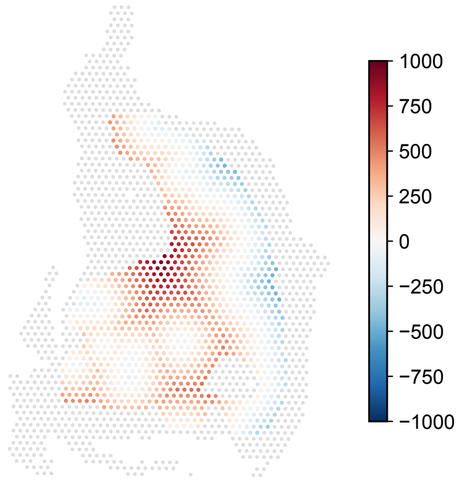
NPJ_sample1 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

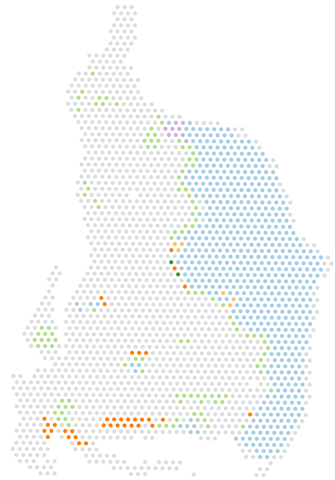
MP10 AUCell and frozen P95



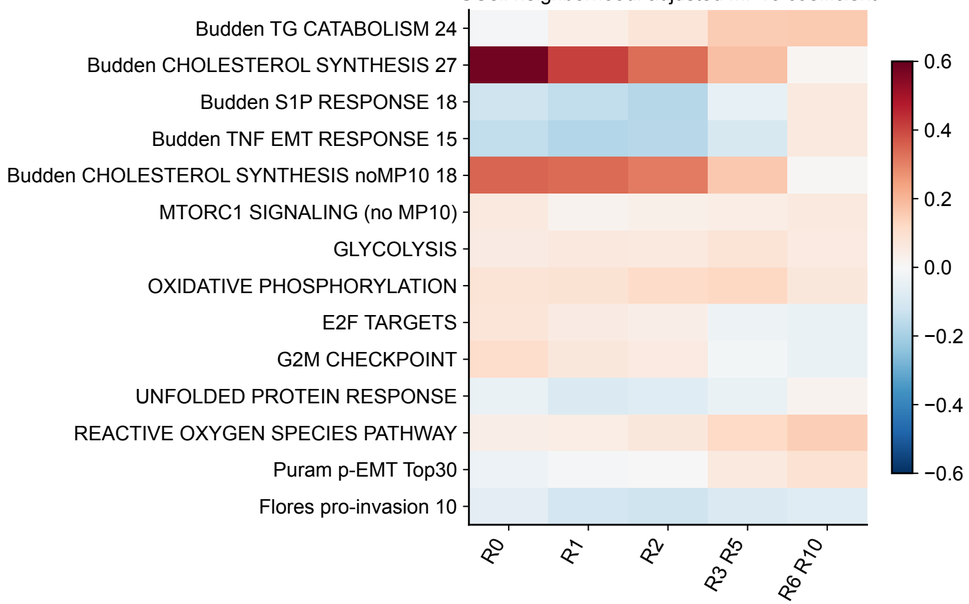
Strict v3.4 signed distance (µm)



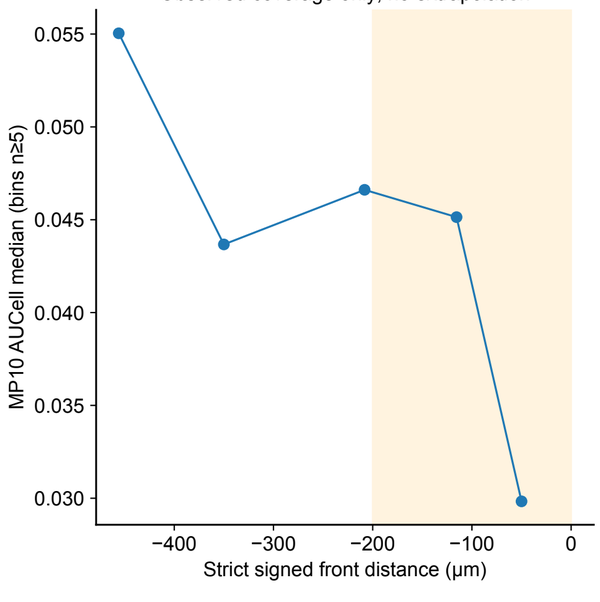
Formal R12 0–8; grey = unlabelled



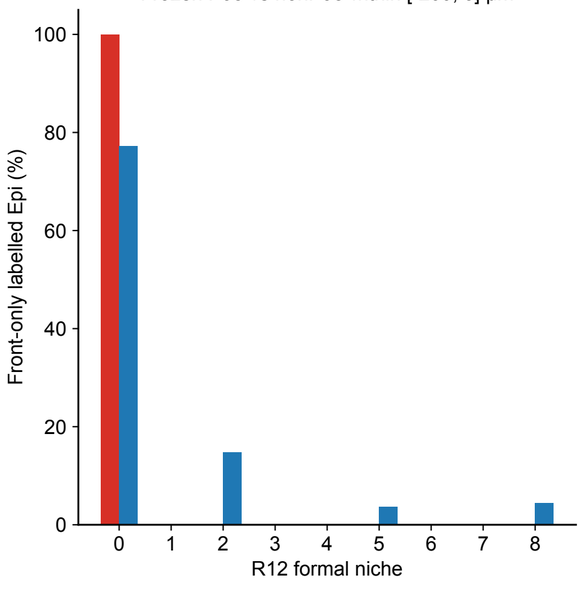
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

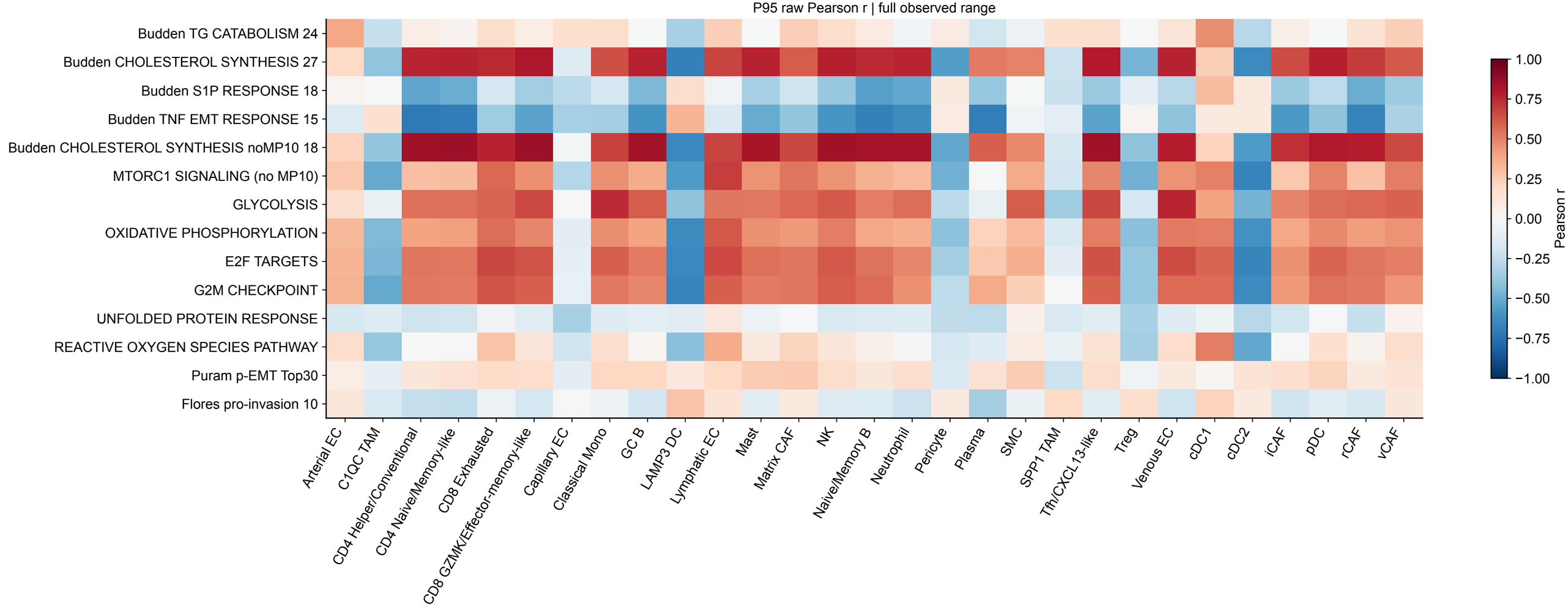


Tissue 2076 | Epi 996 | internal 589 | P95 30
Exact front in Epi 508 | R12 labelled 766

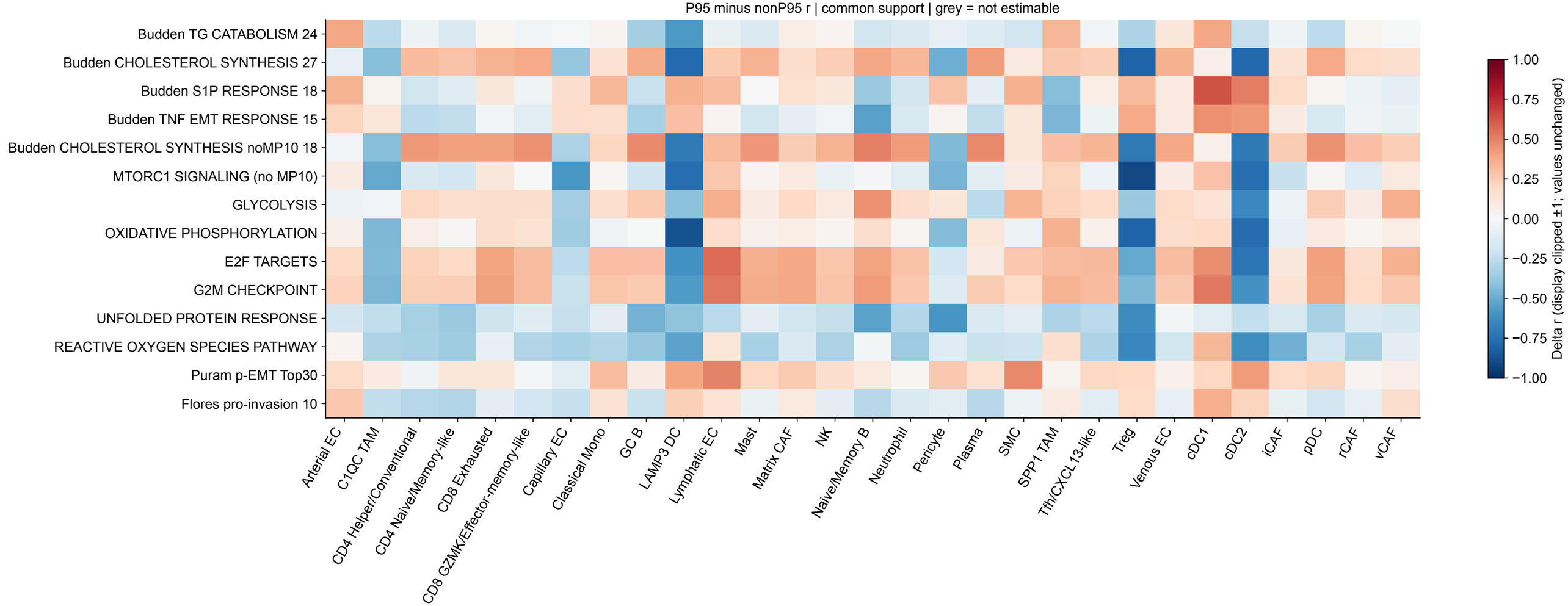
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

NPJ_sample1 | all 29 candidates; fixed core programs

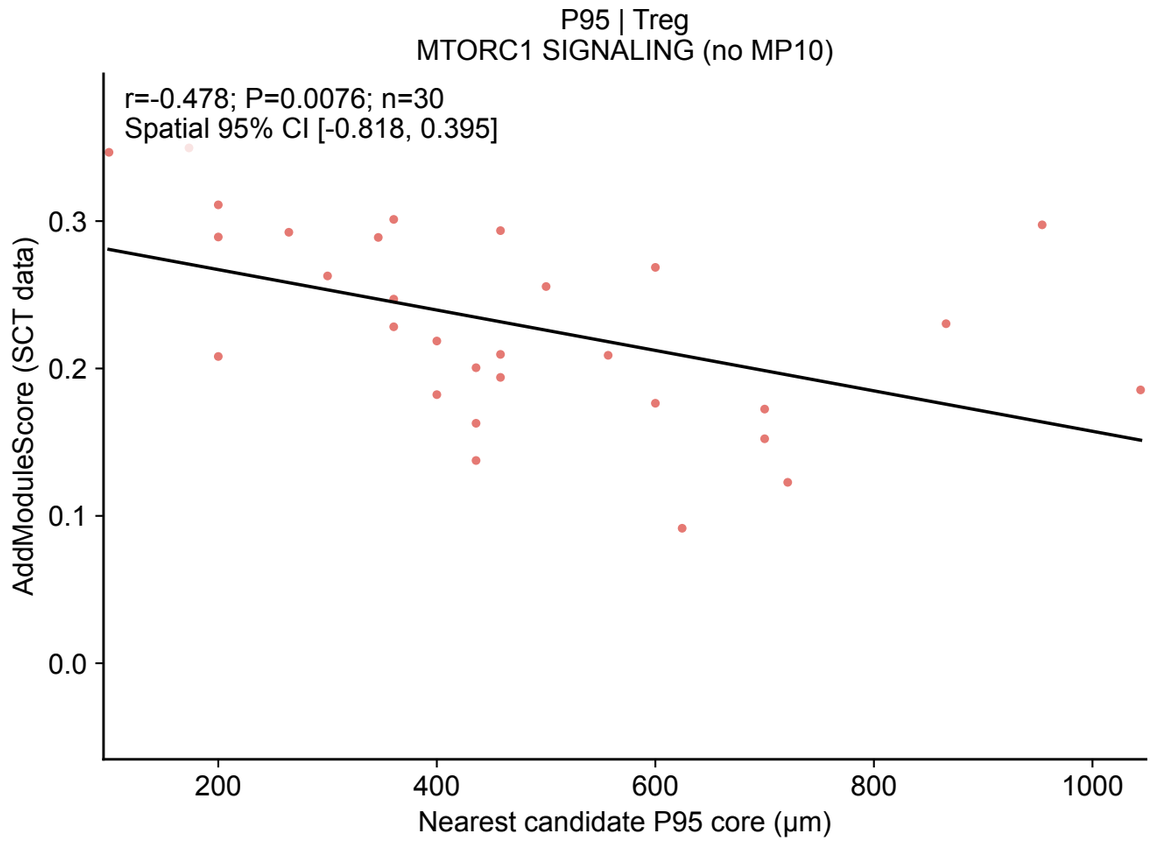
P95 raw Pearson r | full observed range

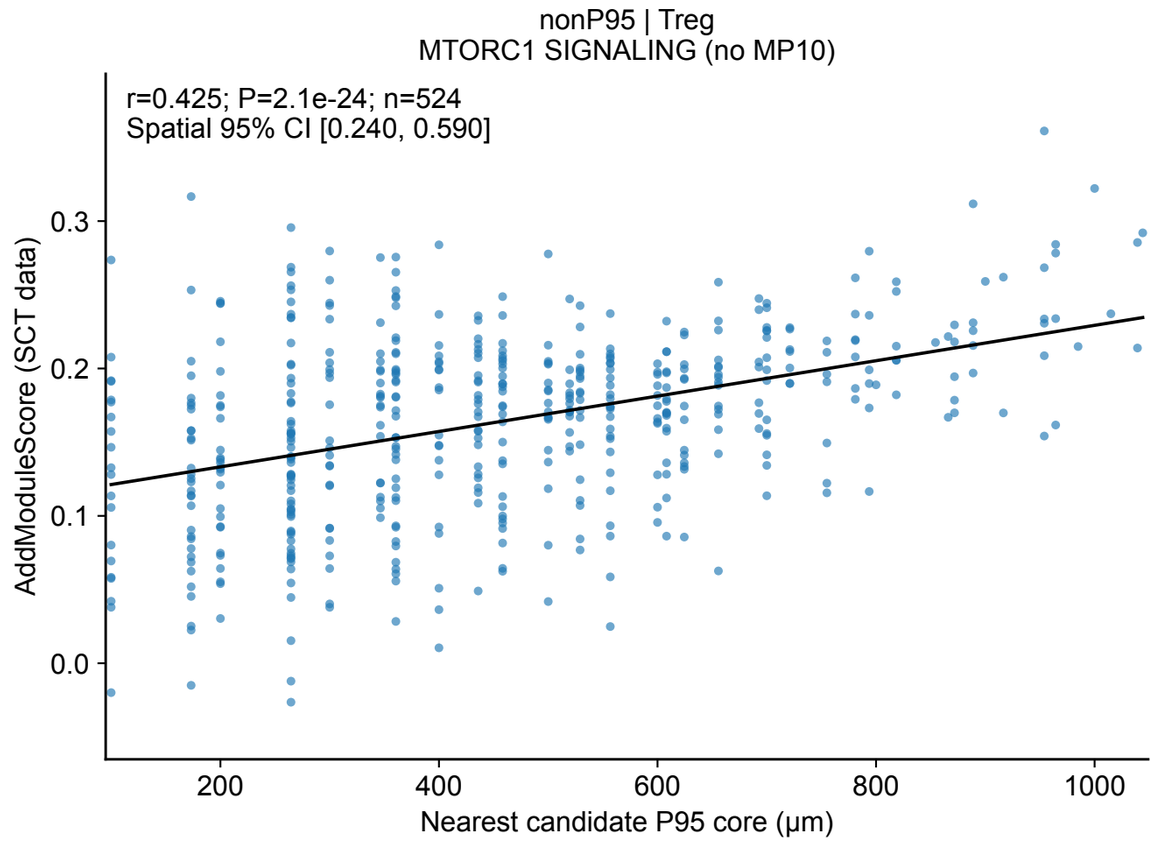


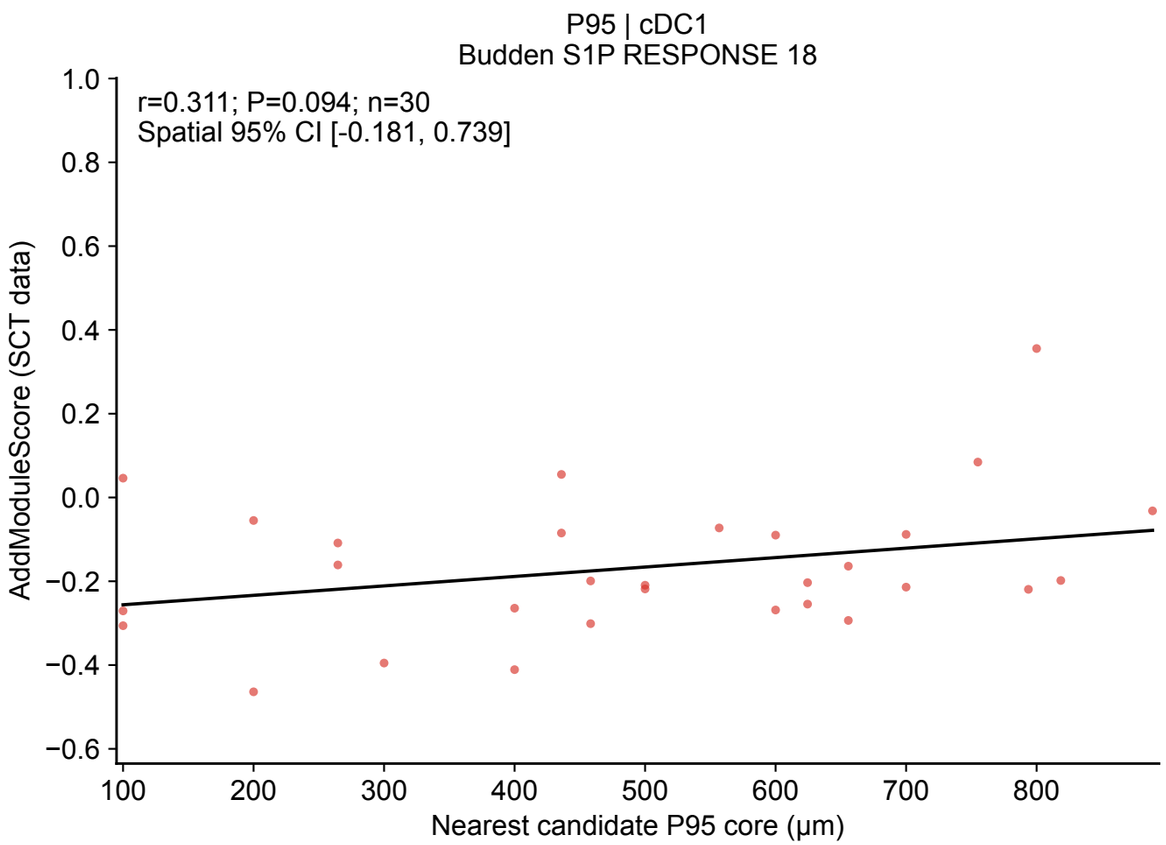
P95 minus nonP95 r | common support | grey = not estimable

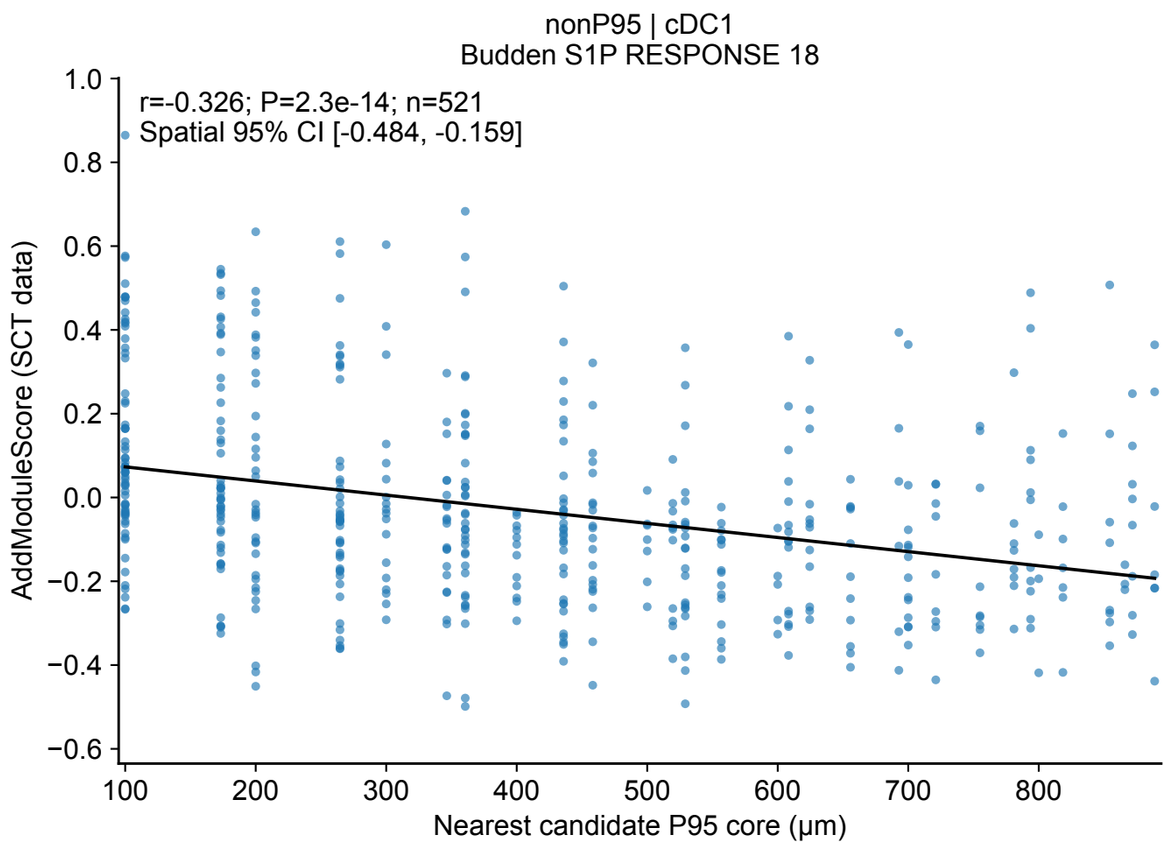


NPJ_sample1 | exploratory scatter display selected by coverage and |delta r|, not P







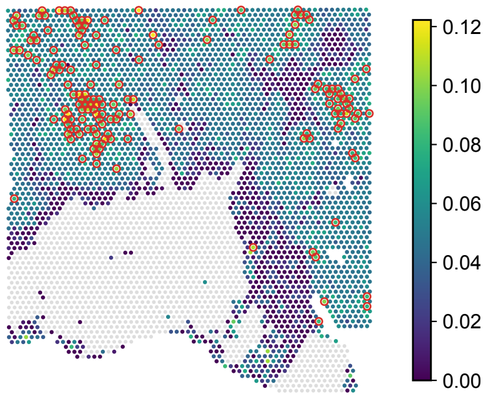


Treg / MTORC1 SIGNALING (no MP10): delta r=-0.903 [-1.232, -0.066]; common 100–1044 µm; spatial P=0.001
cDC1 / Budden S1P RESPONSE 18: delta r=0.637 [0.213, 1.066]; common 100–889 µm; spatial P=0.0023
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

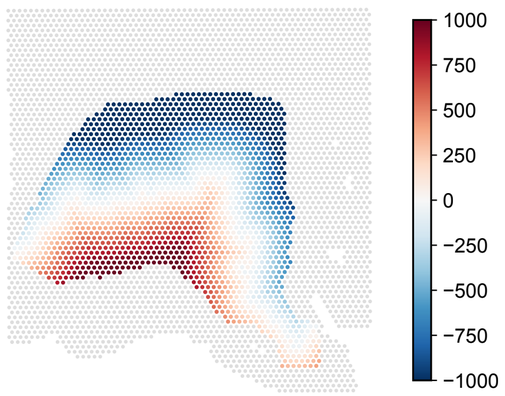
P95 nonP95 distance 0

NPJ_sample2 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

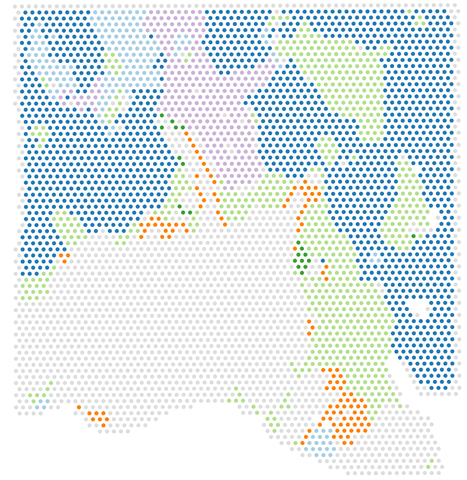
MP10 AUCell and frozen P95



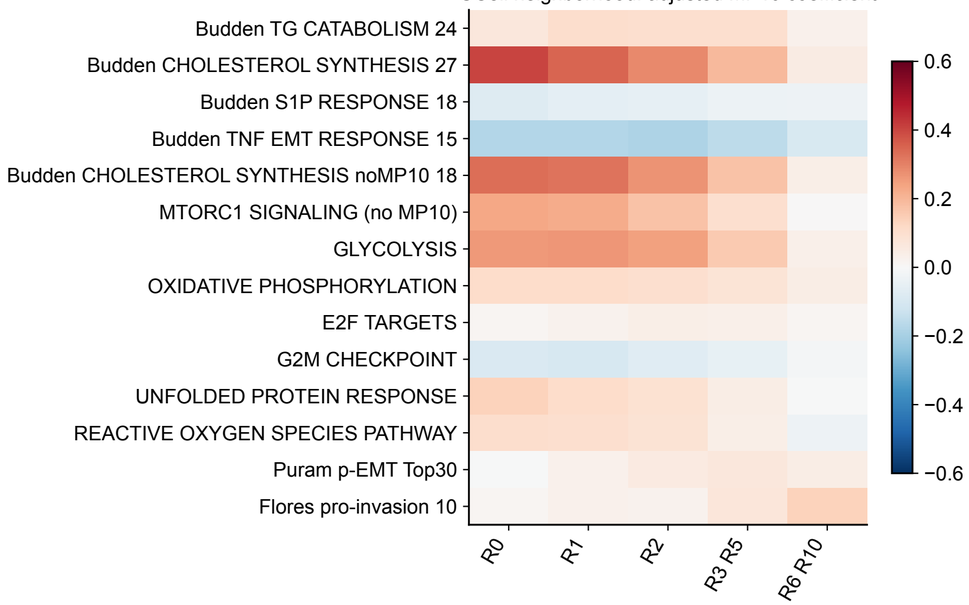
Strict v3.4 signed distance (µm)



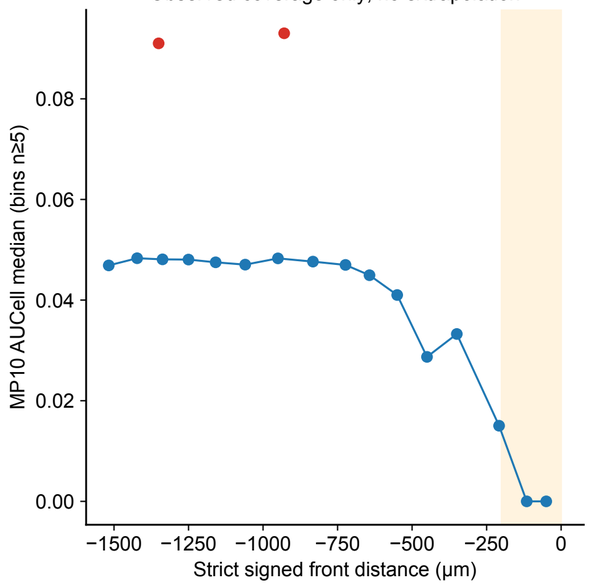
Formal R12 0–8; grey = unlabelled



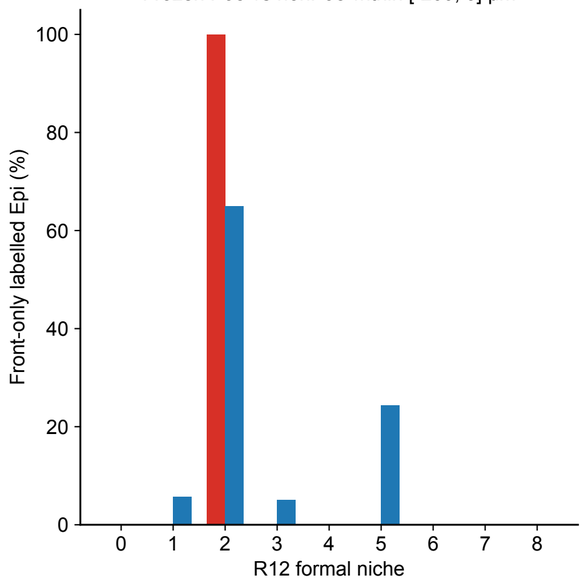
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

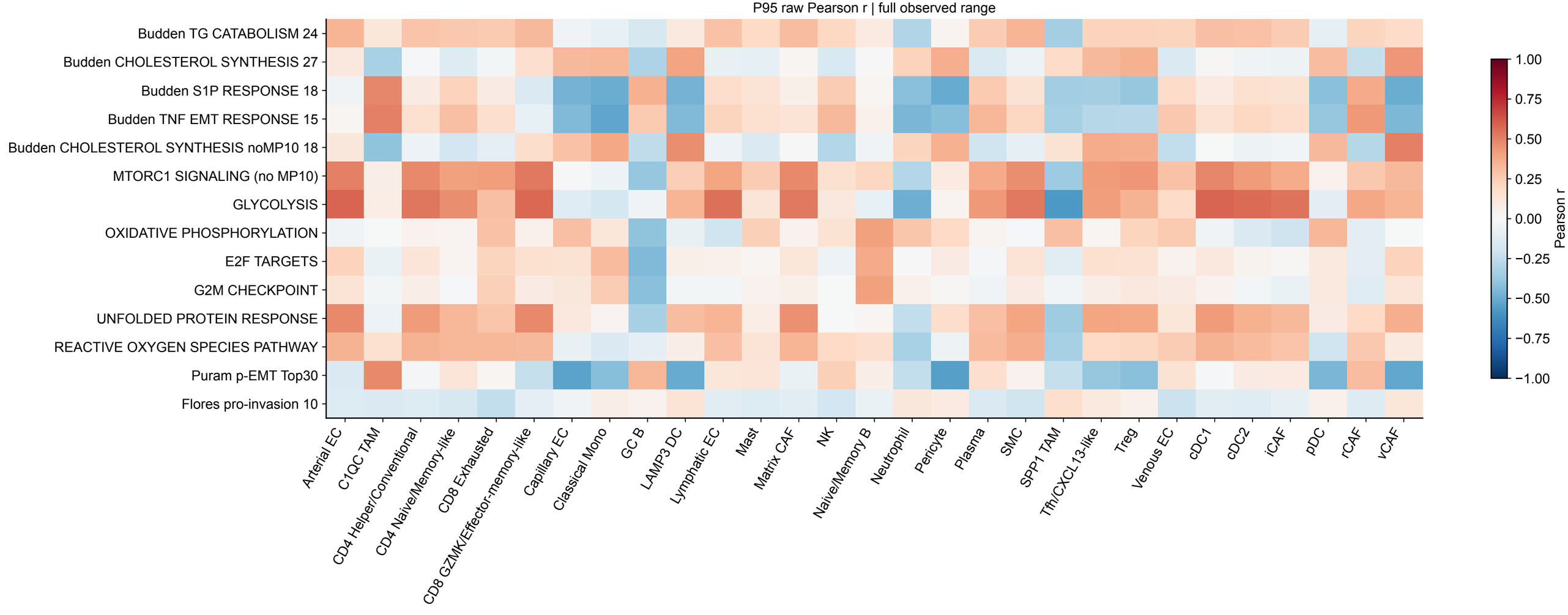


Tissue 4506 | Epi 3217 | internal 3006 | P95 151
Exact front in Epi 1026 | R12 labelled 2832

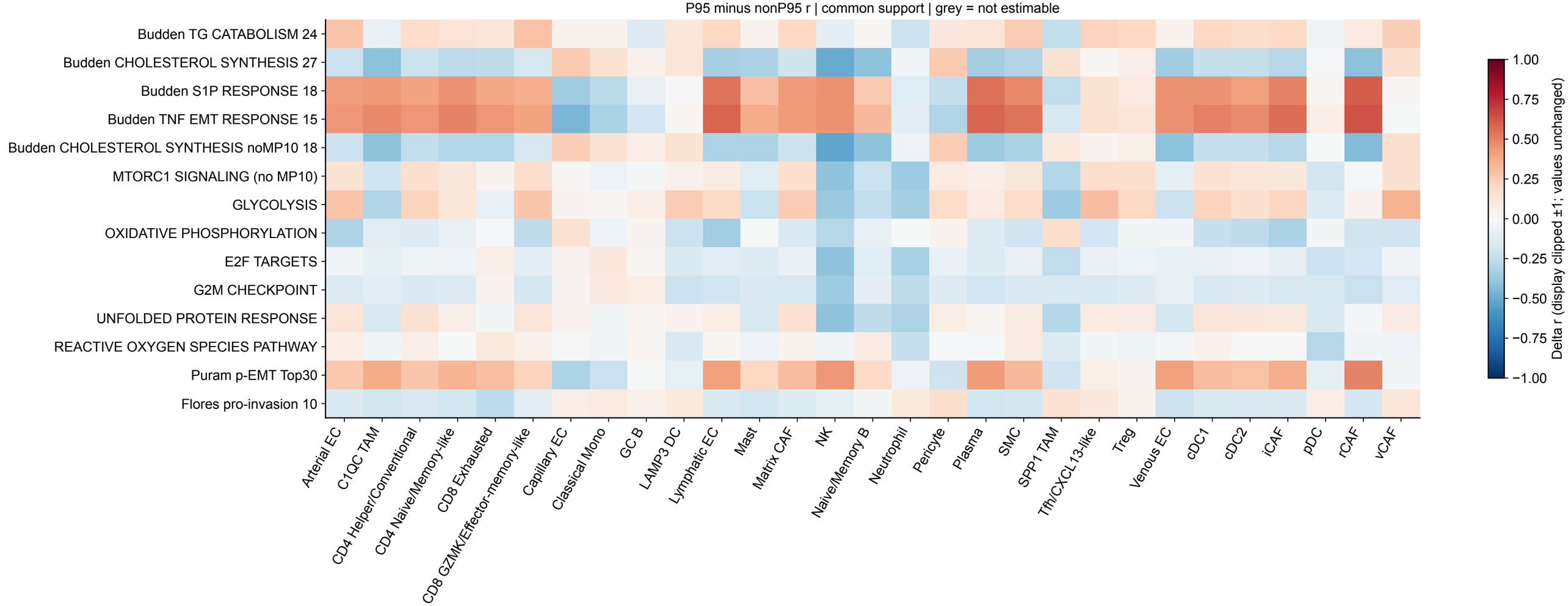
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

NPJ_sample2 | all 29 candidates; fixed core programs

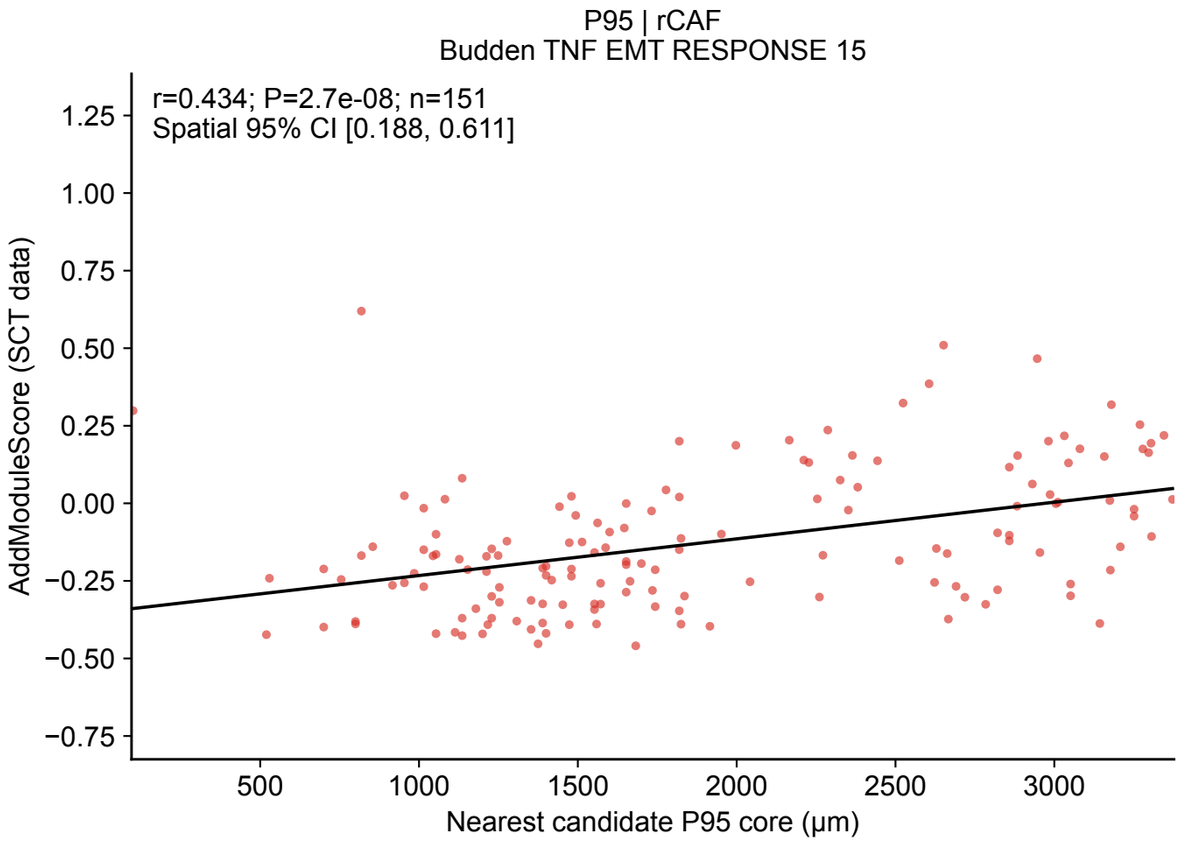
P95 raw Pearson r | full observed range

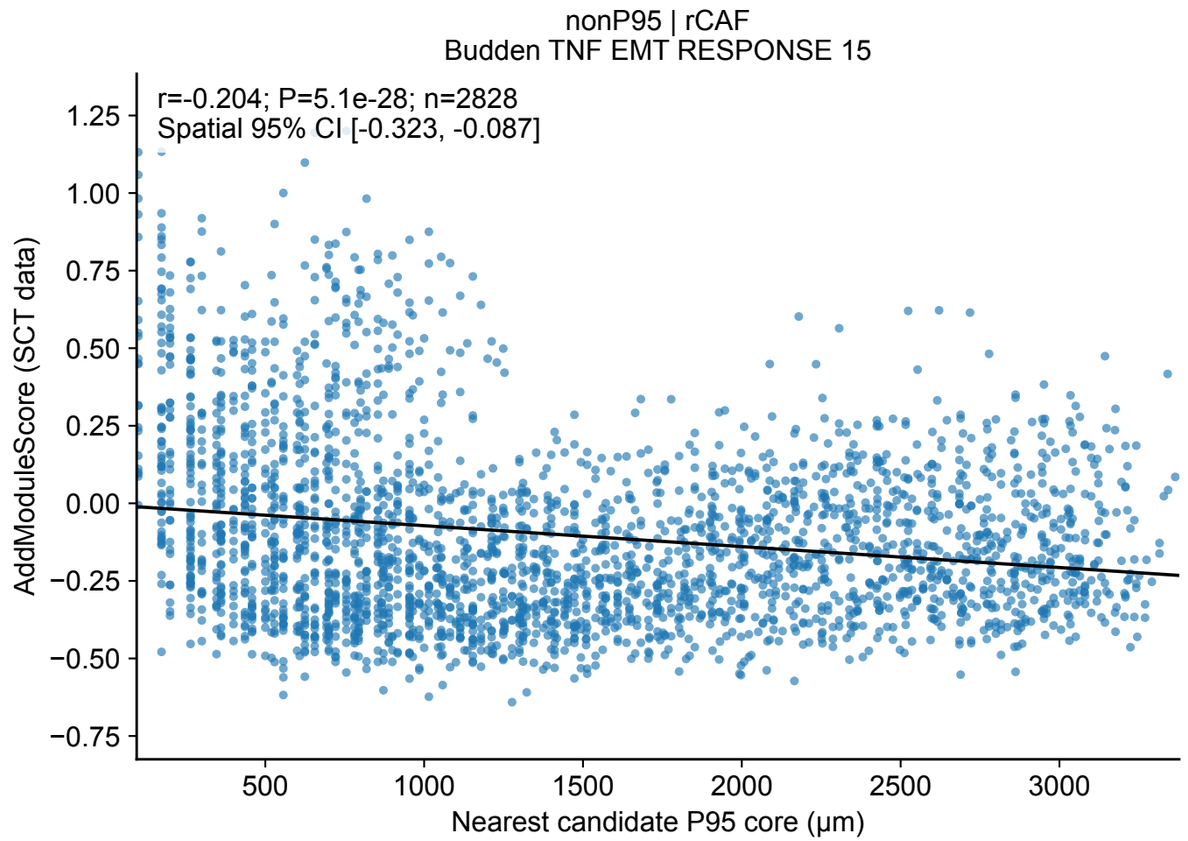


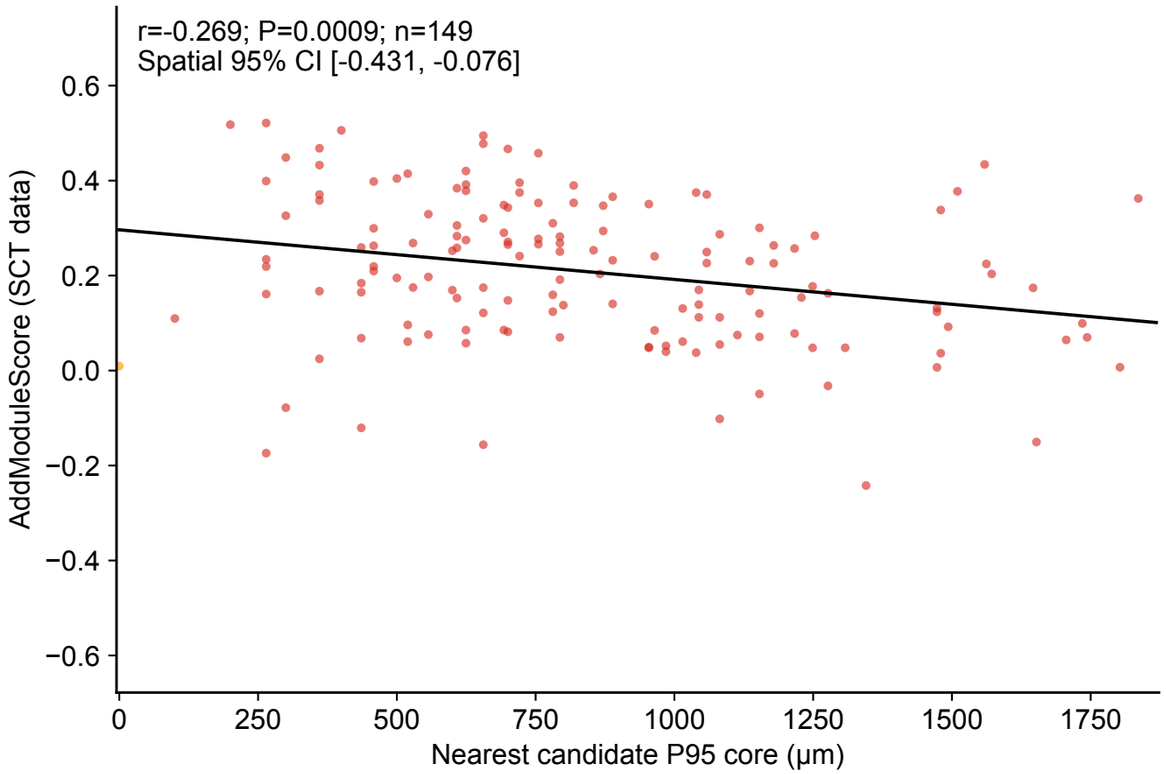
P95 minus nonP95 r | common support | grey = not estimable

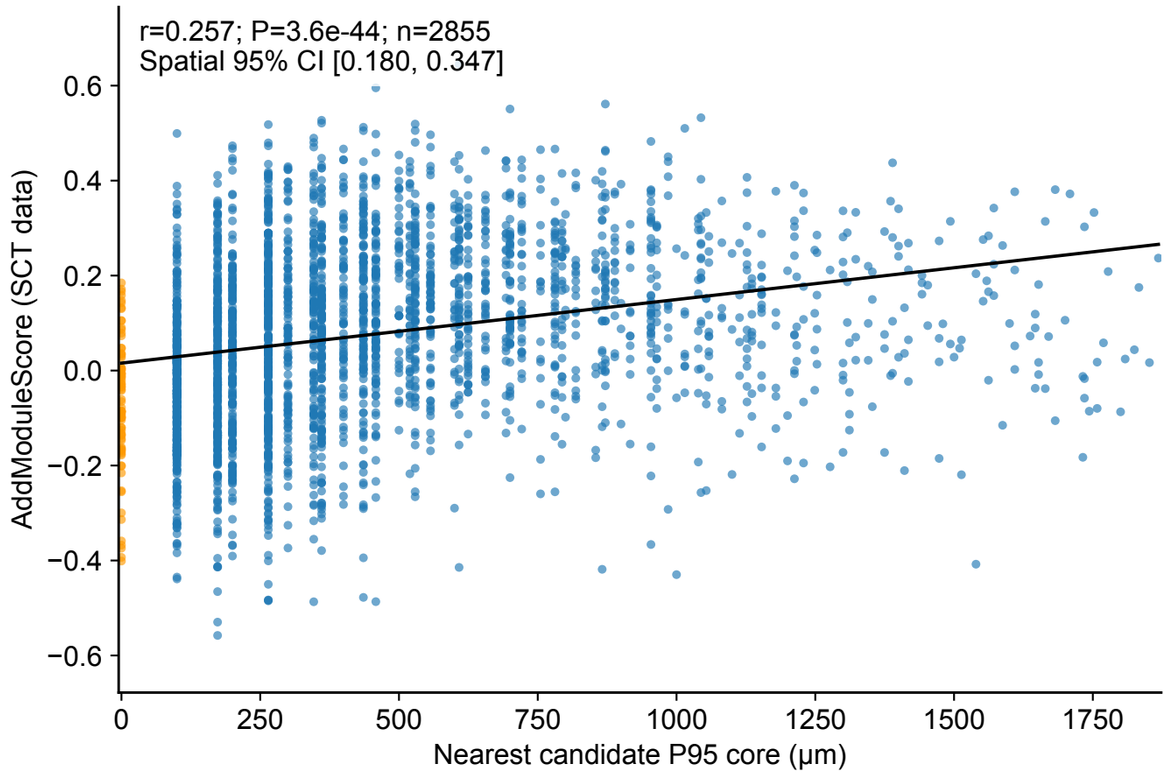


NPJ_sample2 | exploratory scatter display selected by coverage and |delta r|, not P







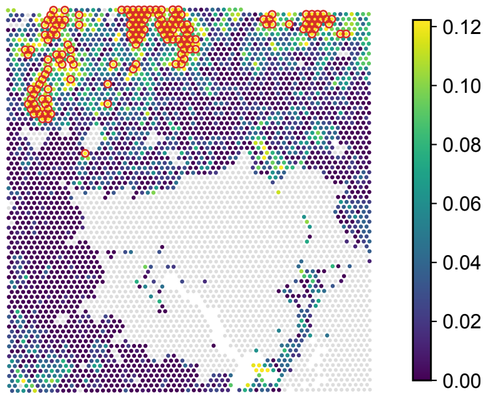


rCAF / Budden TNF EMT RESPONSE 15: delta r=0.638 [0.411, 0.822]; common 100–3372 µm; spatial P=4.7e-07
NK / Budden CHOLESTEROL SYNTHESIS noMP10 18: delta r=-0.526 [-0.679, -0.347]; common 0–1868 µm; spatial P=5.5e-09
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

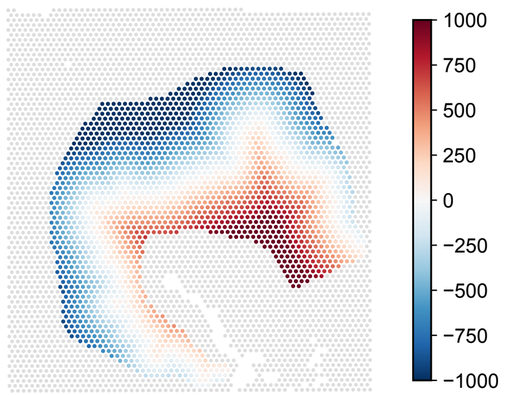
P95 nonP95 distance 0

NPJ_sample3 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

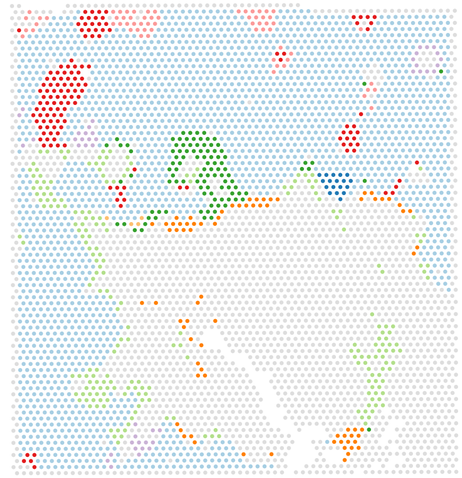
MP10 AUCell and frozen P95



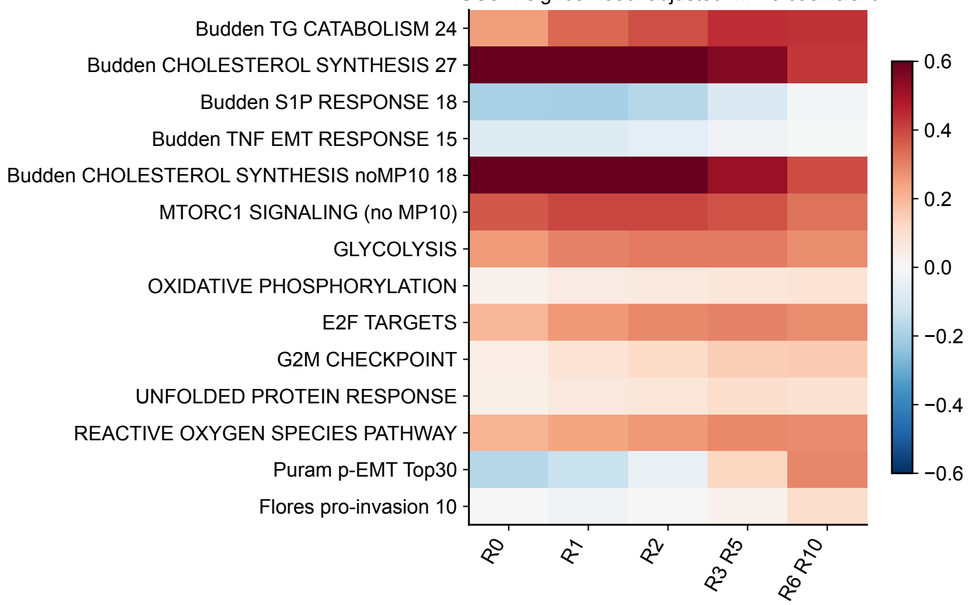
Strict v3.4 signed distance (µm)



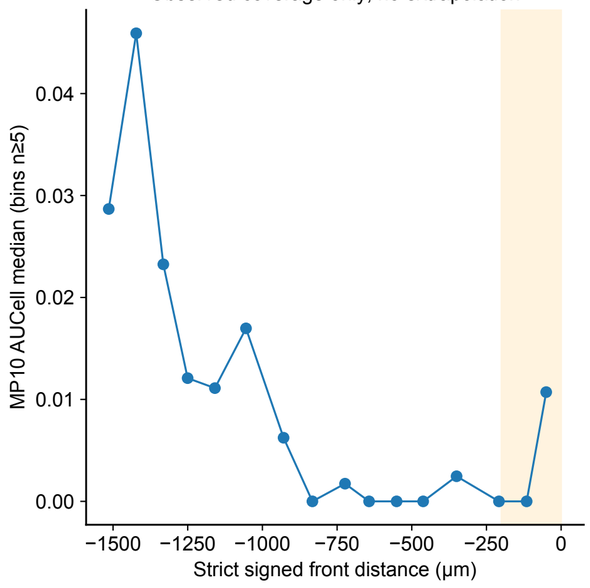
Formal R12 0–8; grey = unlabelled



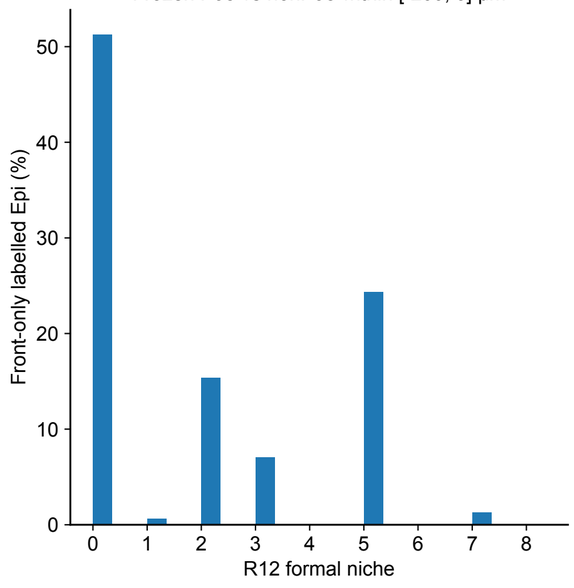
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

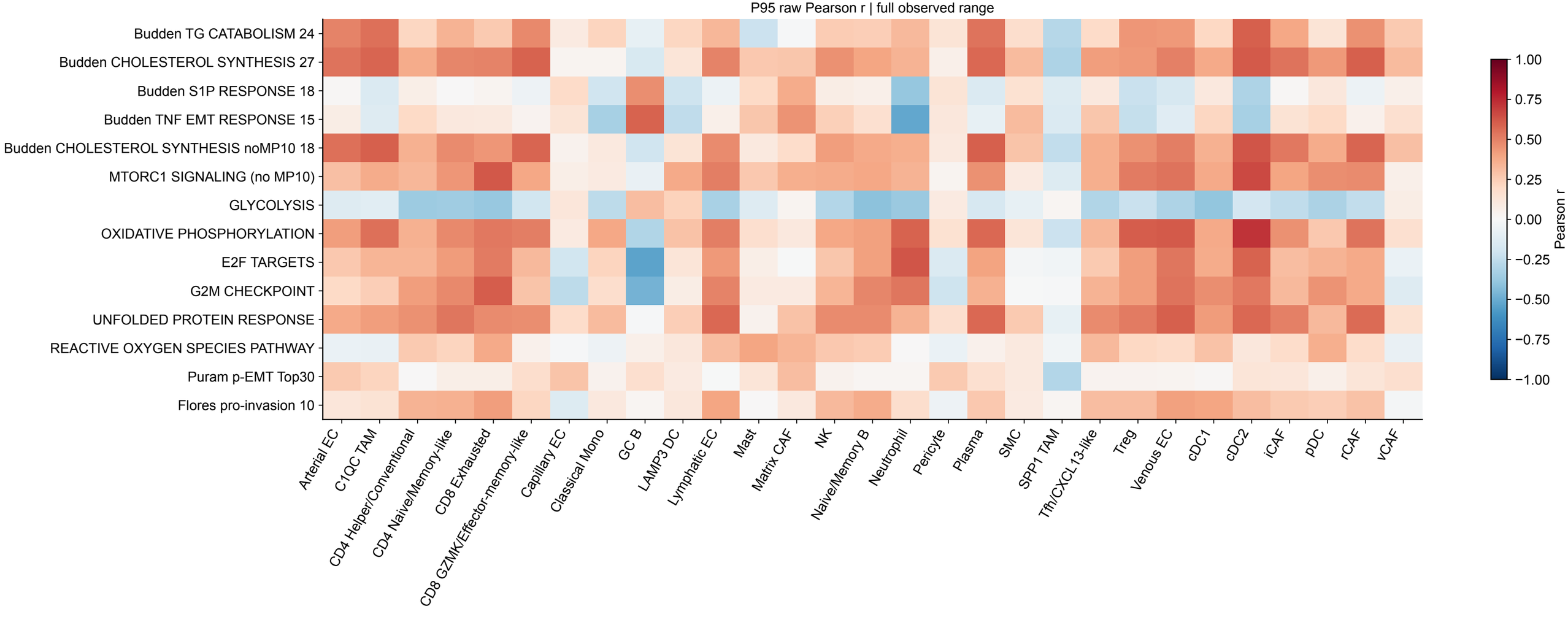


Tissue 4881 | Epi 3182 | internal 2955 | P95 148
Exact front in Epi 1207 | R12 labelled 2767

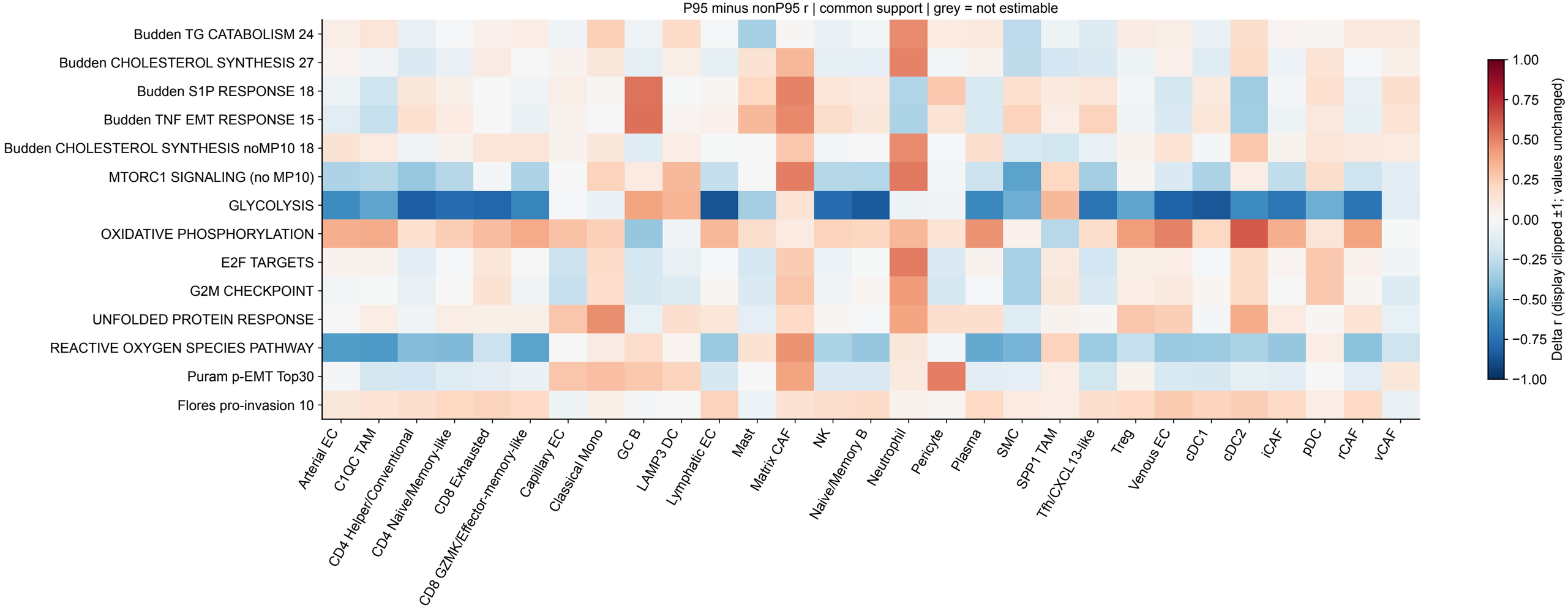


NPJ_sample3 | all 29 candidates; fixed core programs

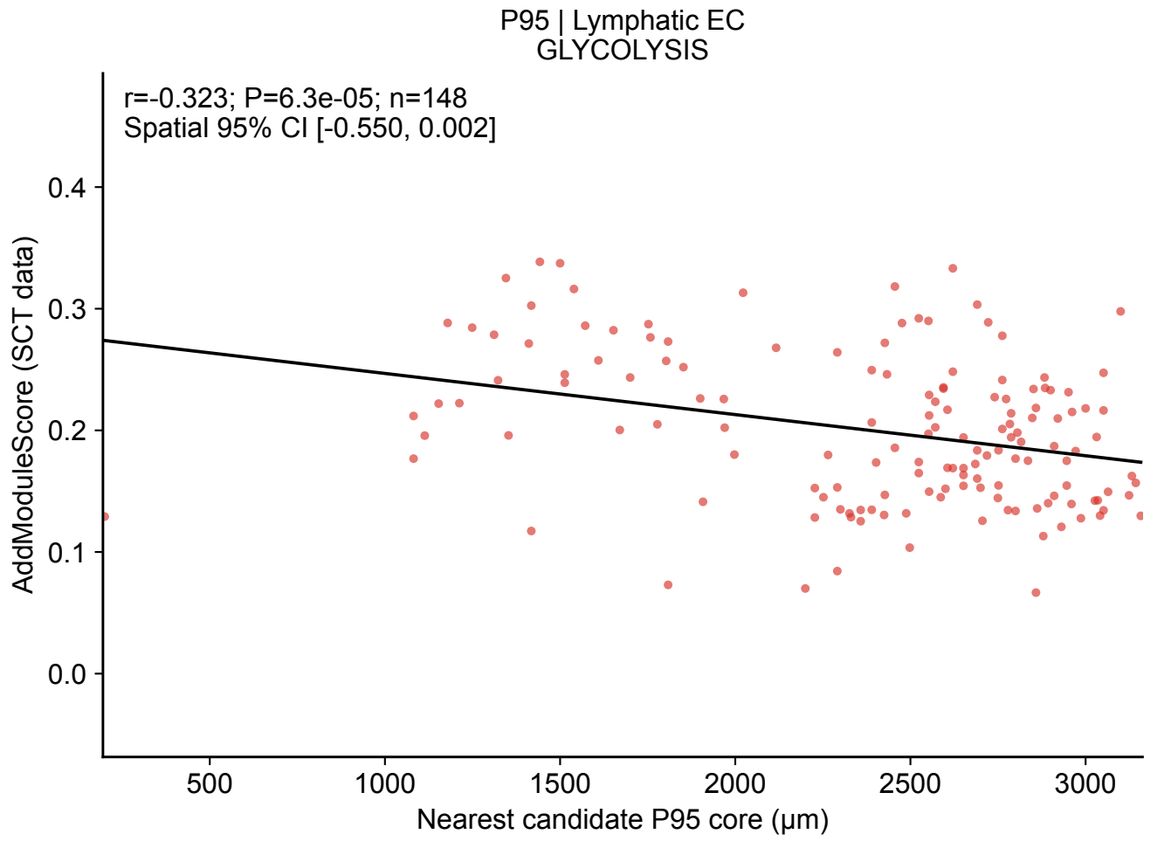
P95 raw Pearson r | full observed range

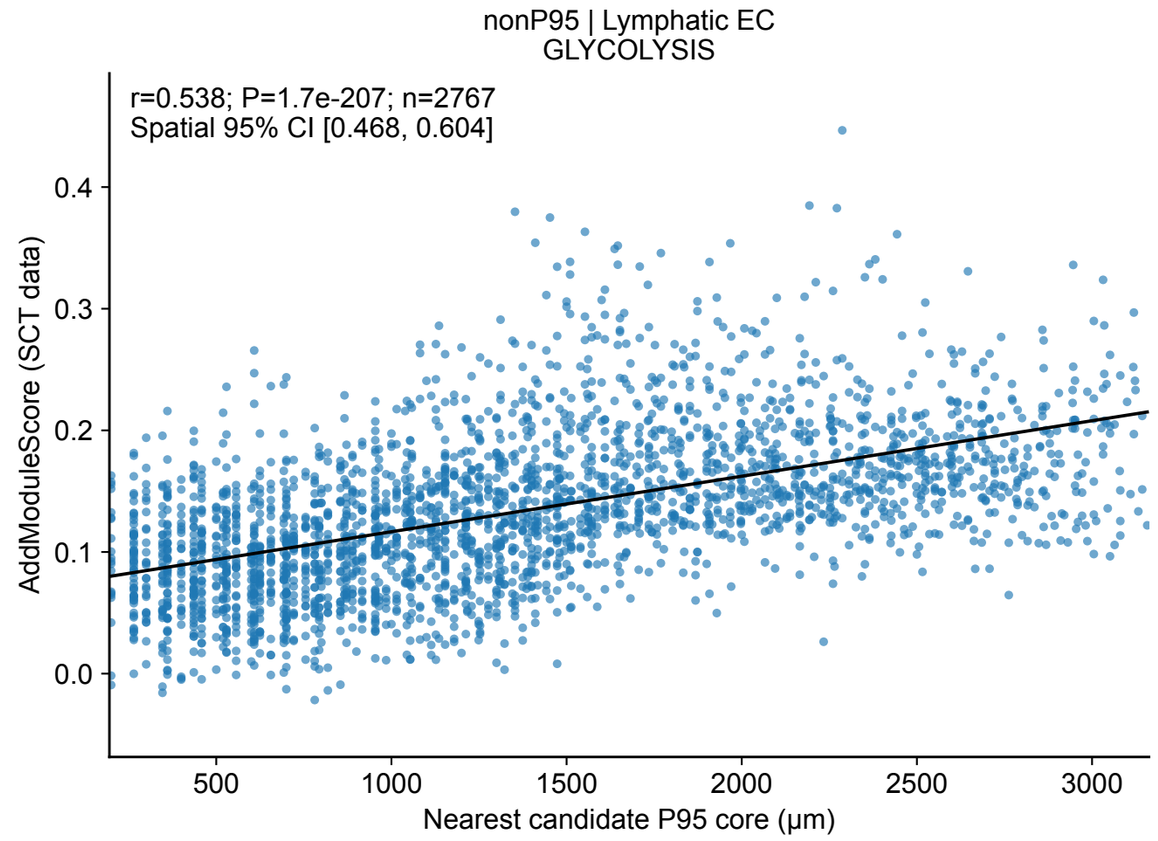


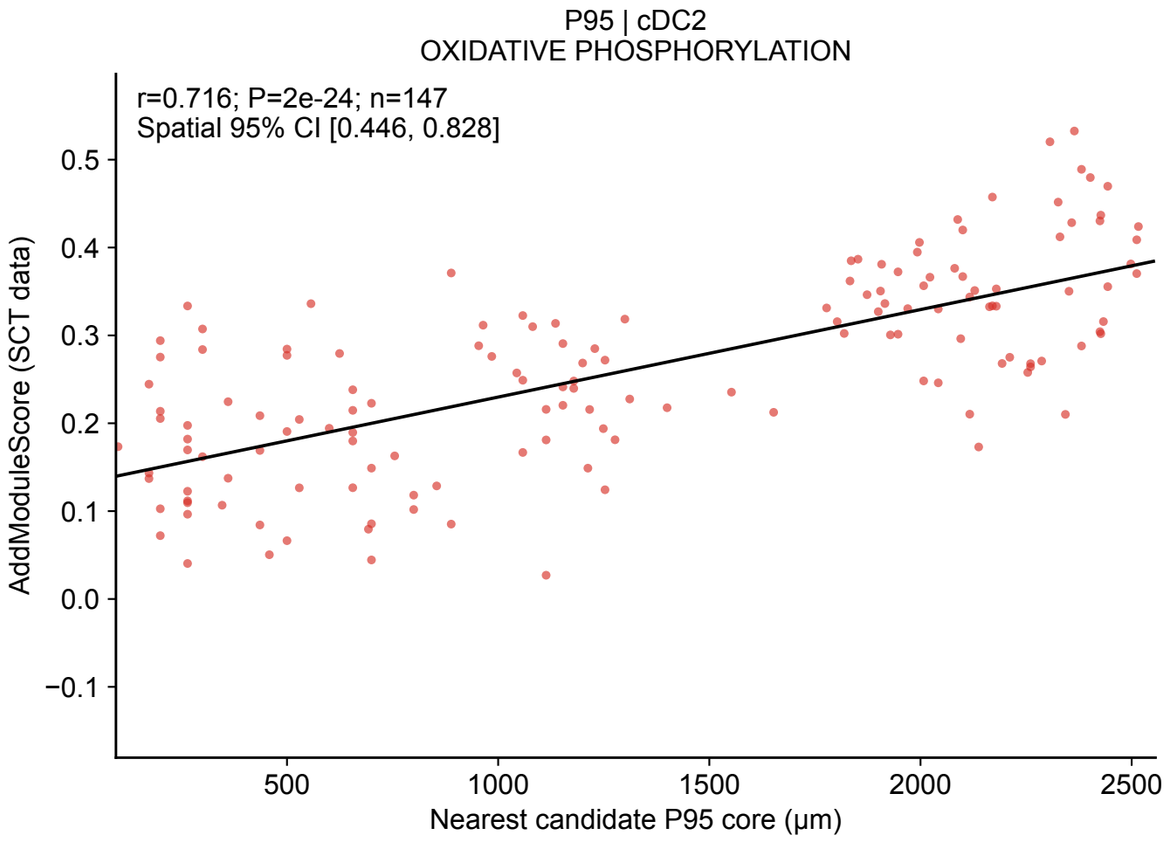
P95 minus nonP95 r | common support | grey = not estimable

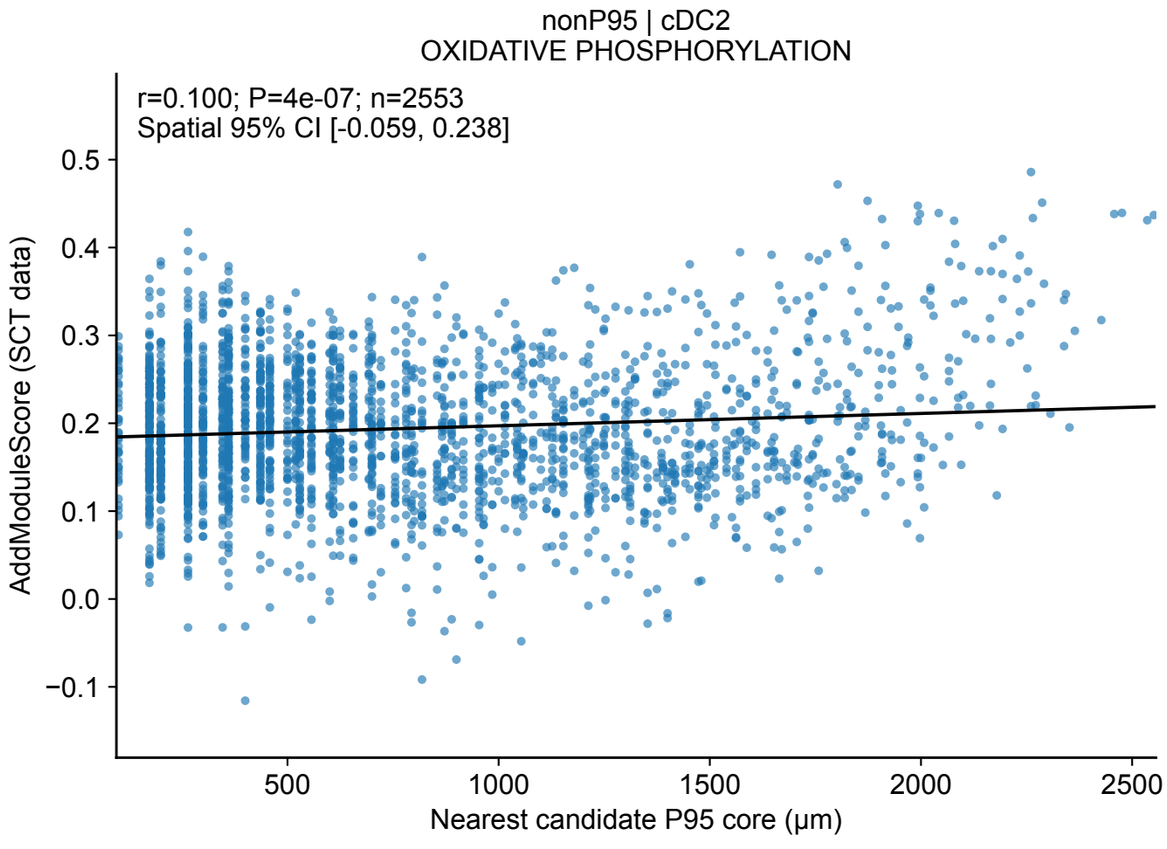


NPJ_sample3 | exploratory scatter display selected by coverage and |delta r|, not P





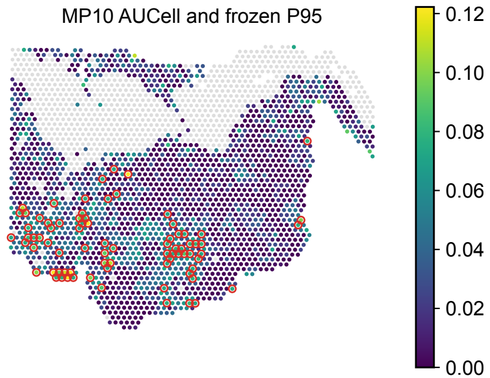


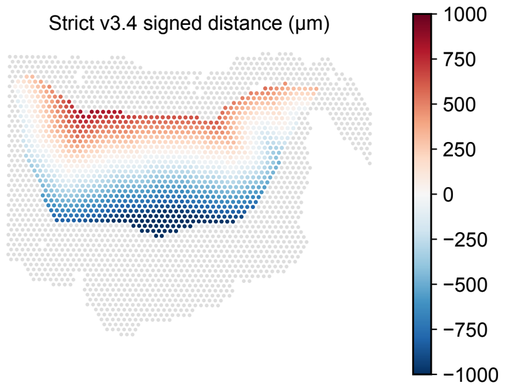


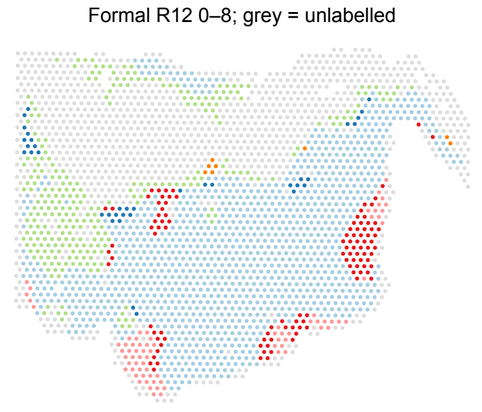
Lymphatic EC / GLYCOLYSIS: delta r=-0.861 [-1.073, -0.534]; common 200–3158 μm; spatial P=2.4e-09
cDC2 / OXIDATIVE PHOSPHORYLATION: delta r=0.616 [0.375, 0.800]; common 100–2551 μm; spatial P=1.1e-06
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 μm blocks.

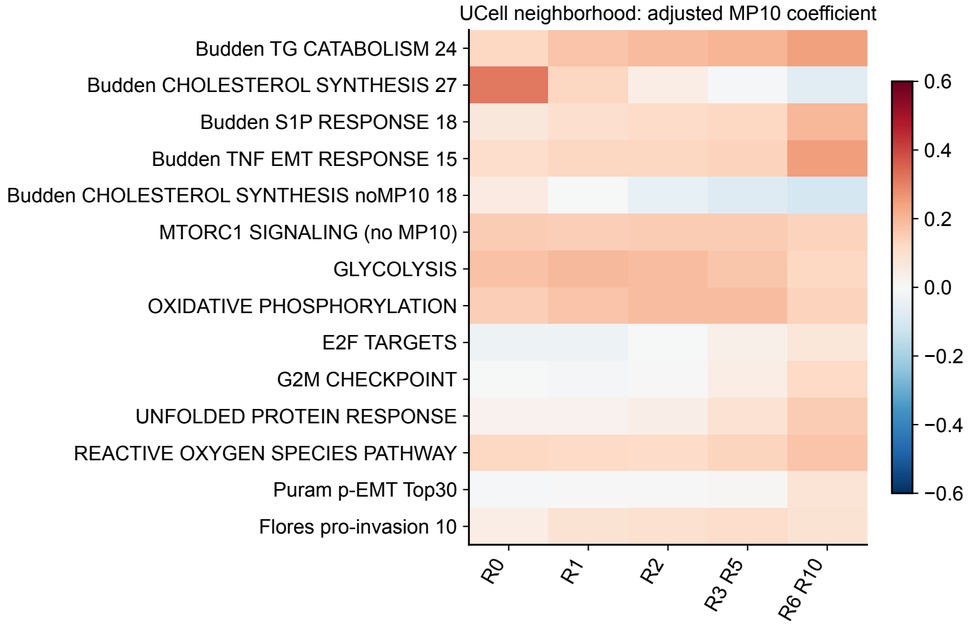
P95 nonP95 distance 0

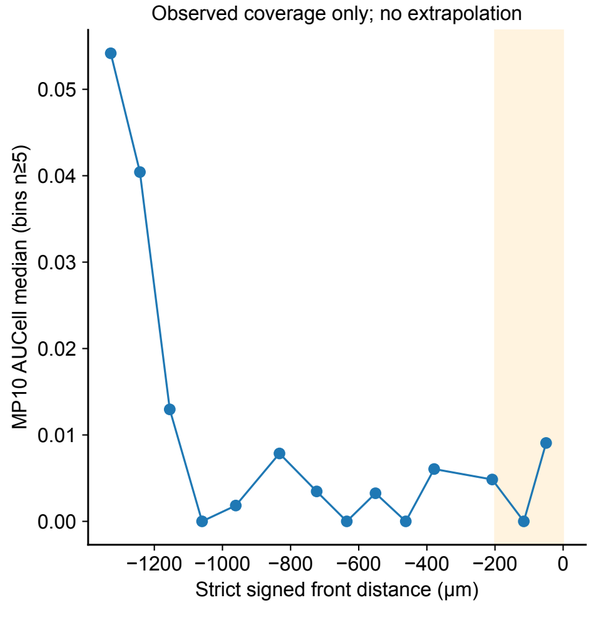
NPJ_sample4 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

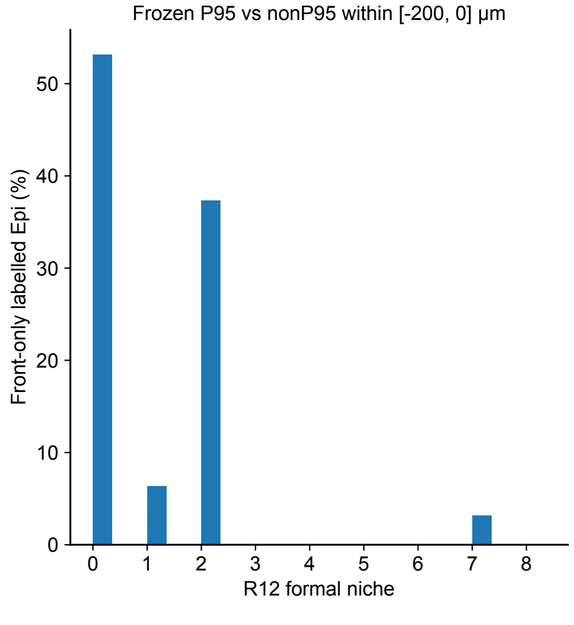










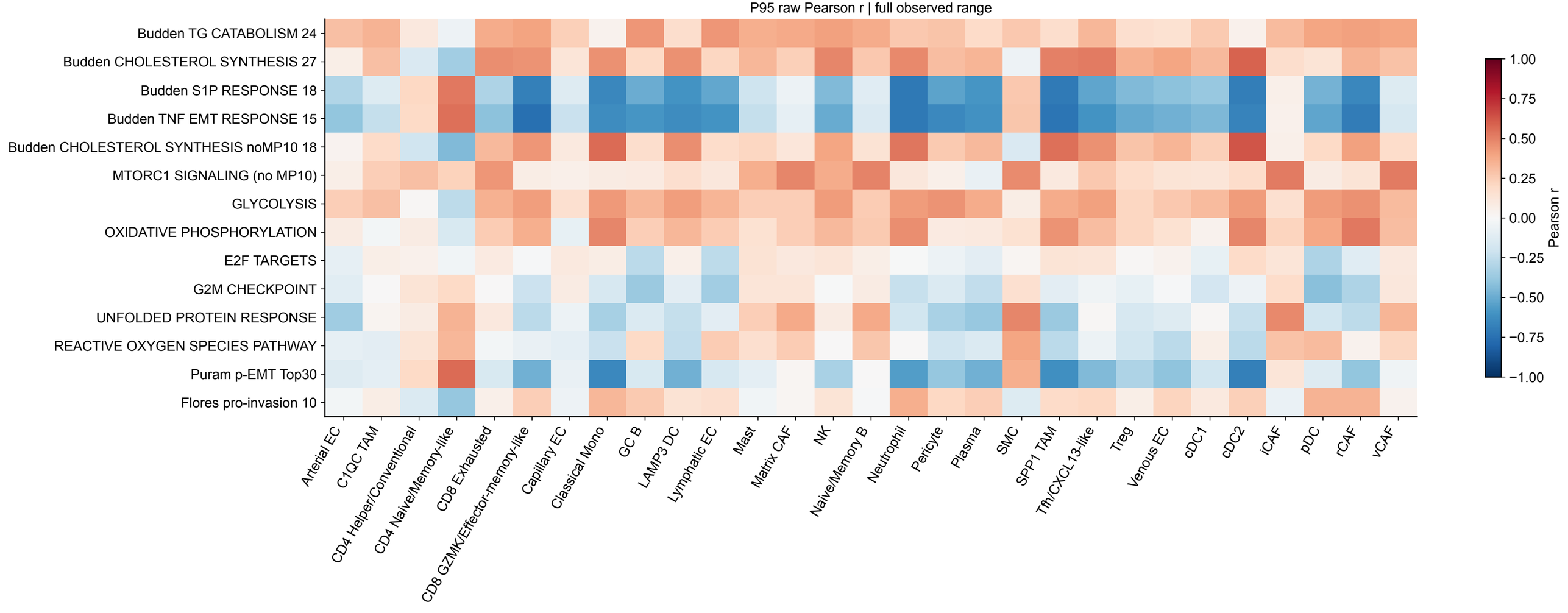


Tissue 2639 | Epi 1885 | internal 1569 | P95 79
Exact front in Epi 701 | R12 labelled 1655

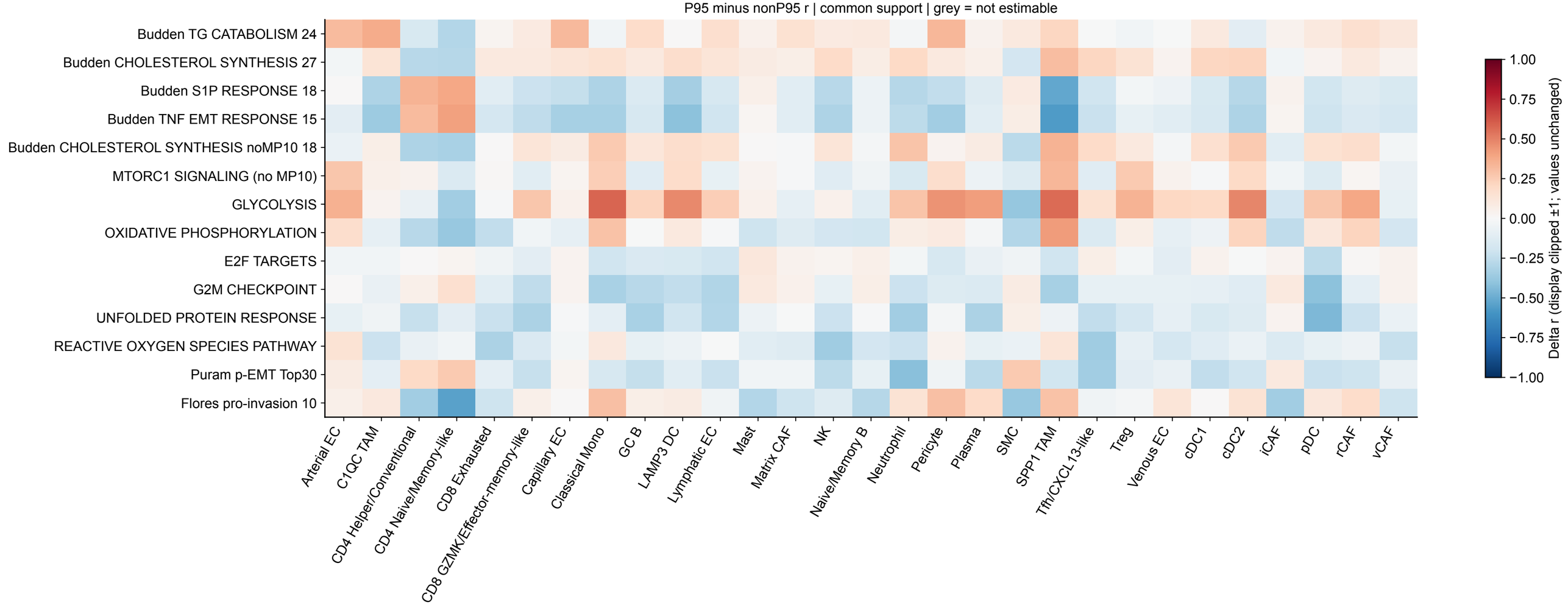
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

NPJ_sample4 | all 29 candidates; fixed core programs

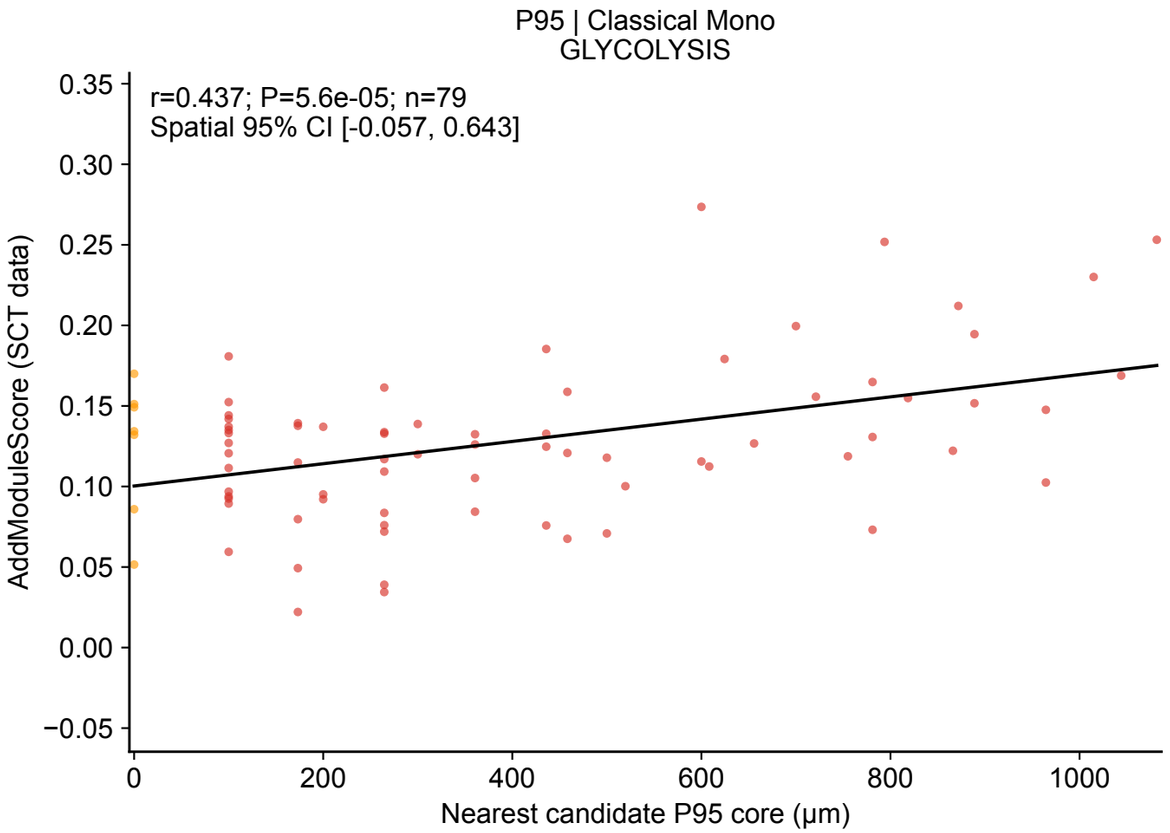
P95 raw Pearson r | full observed range

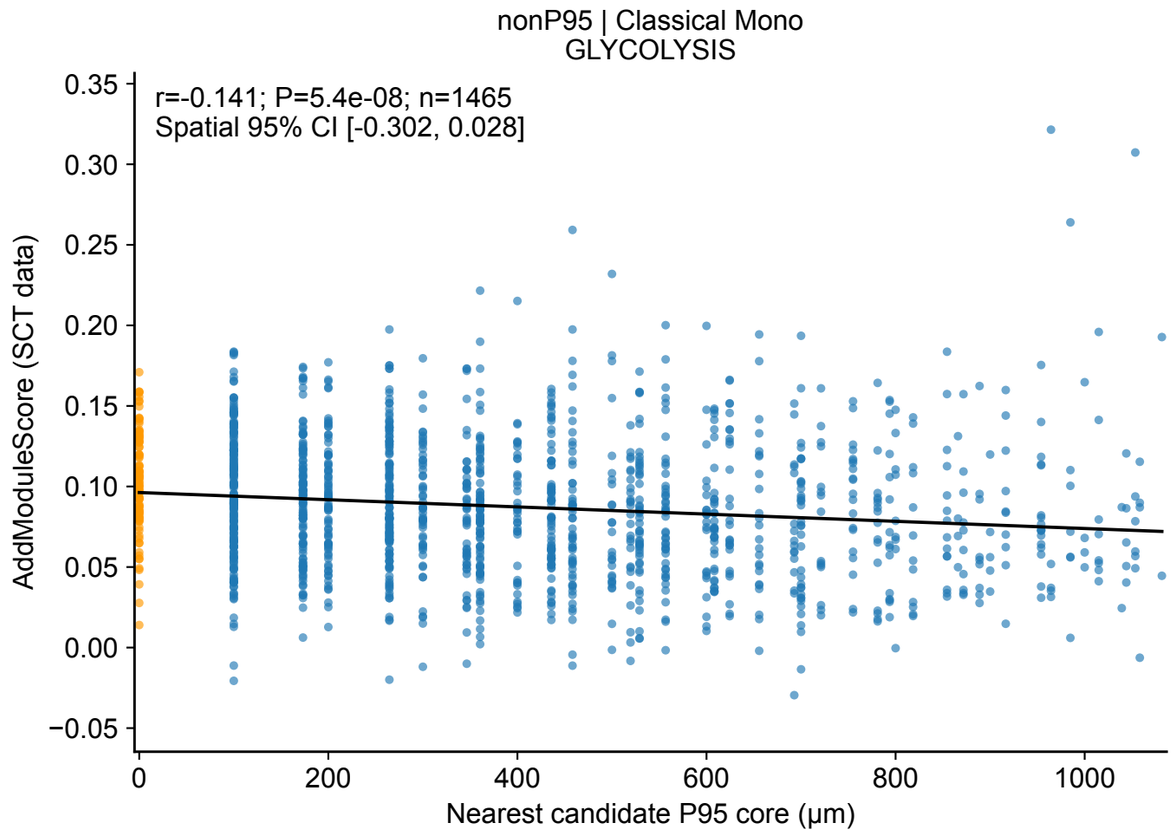


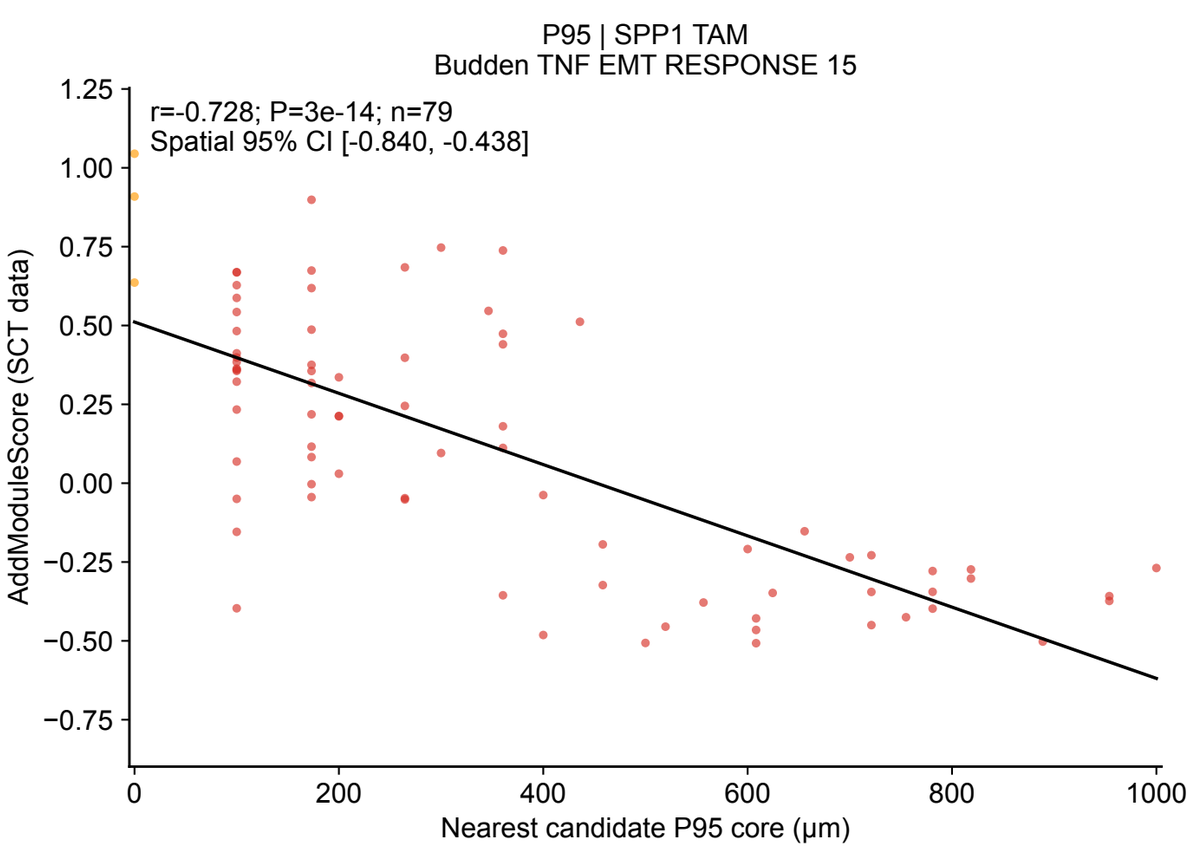
P95 minus nonP95 r | common support | grey = not estimable

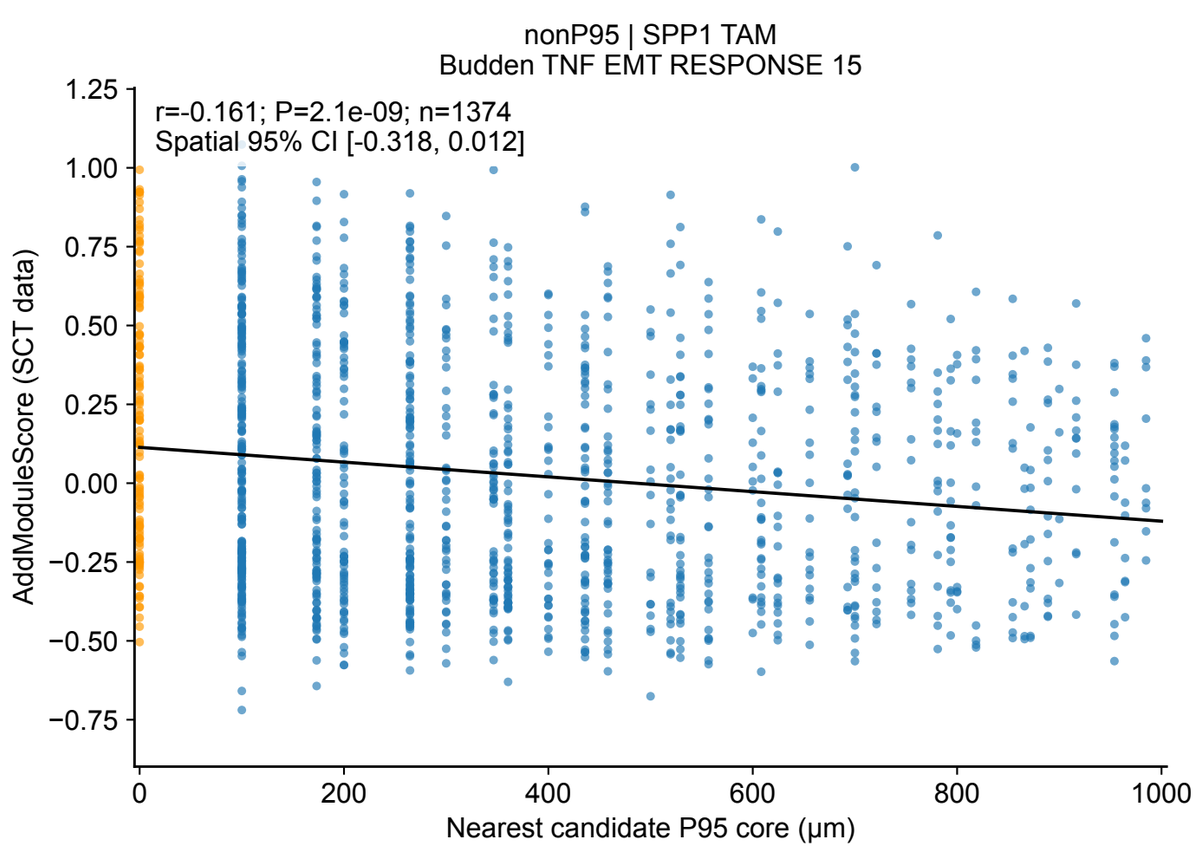


NPJ_sample4 | exploratory scatter display selected by coverage and |delta r|, not P





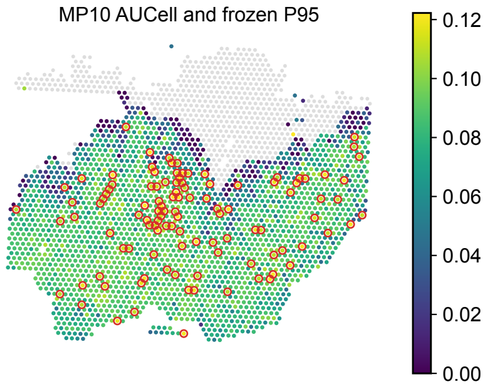


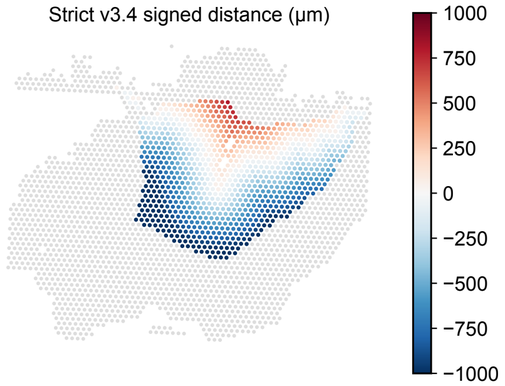


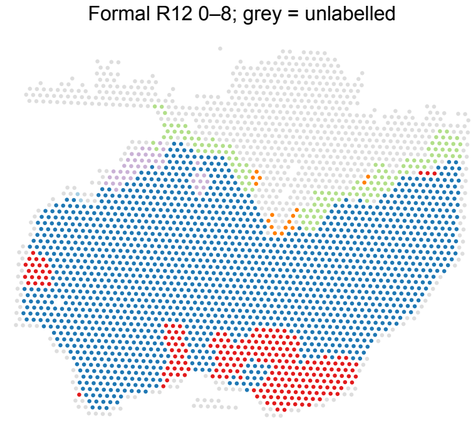
Classical Mono / GLYCOLYSIS: delta r=0.579 [0.186, 0.742]; common 0–1082 μm; spatial P=5.5e-06
SPP1 TAM / Budden TNF EMT RESPONSE 15: delta r=-0.567 [-0.746, -0.306]; common 0–1000 μm; spatial P=5.8e-06
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 μm blocks.

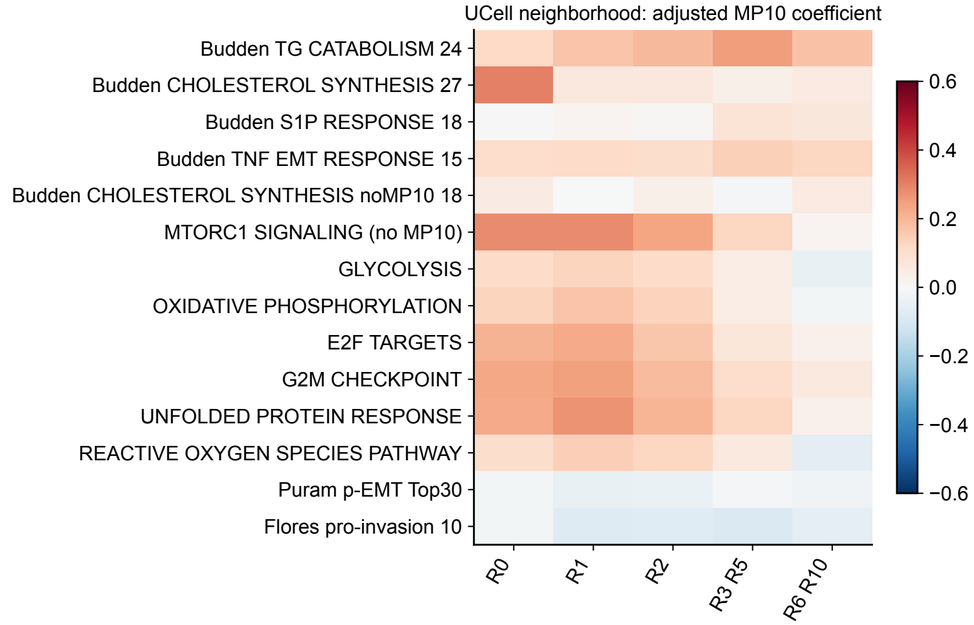
P95 nonP95 distance 0

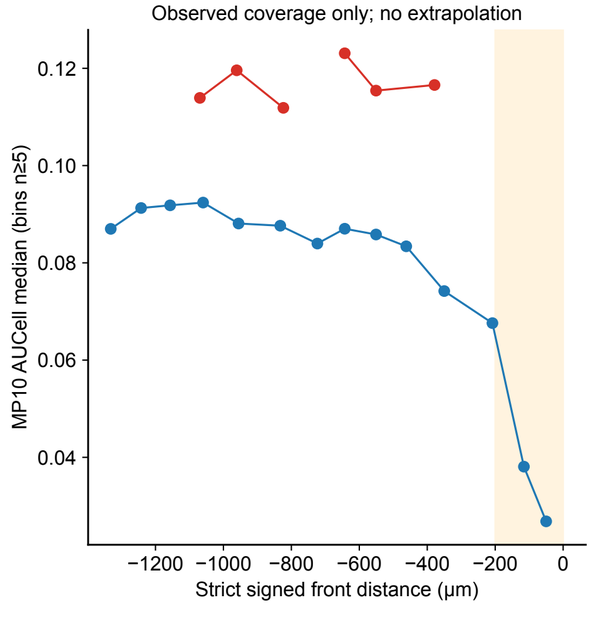
NPJ_sample5 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

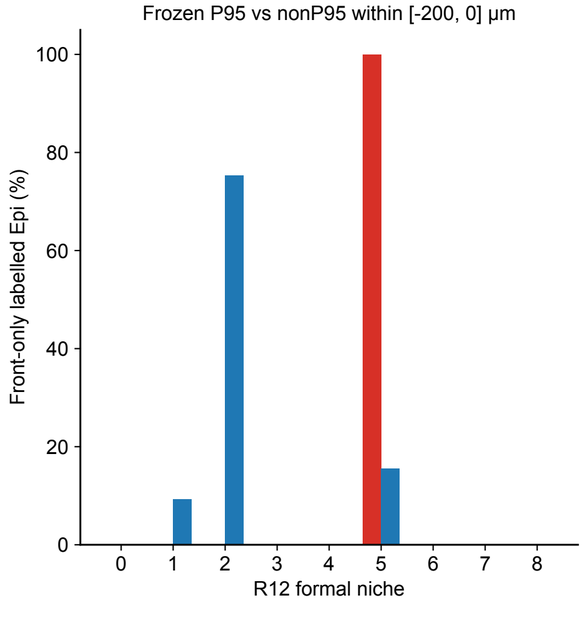










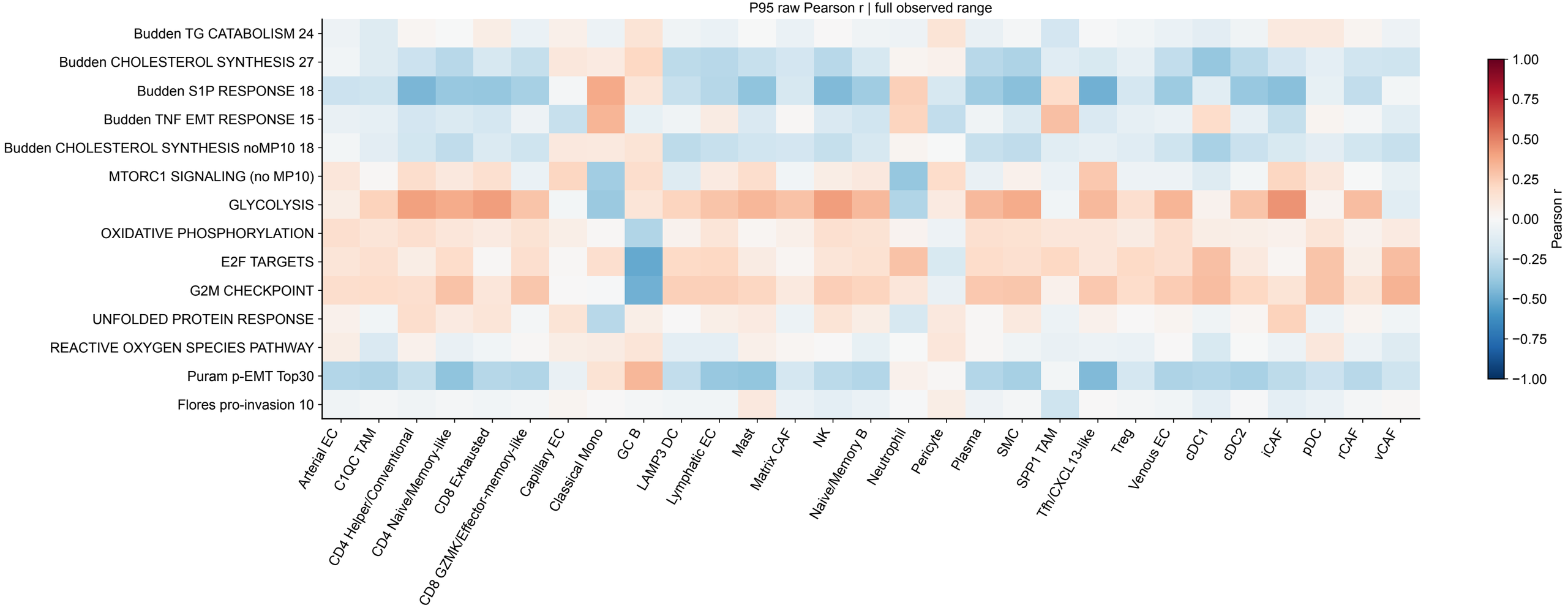


Tissue 2815 | Epi 2186 | internal 2178 | P95 109
Exact front in Epi 656 | R12 labelled 2006

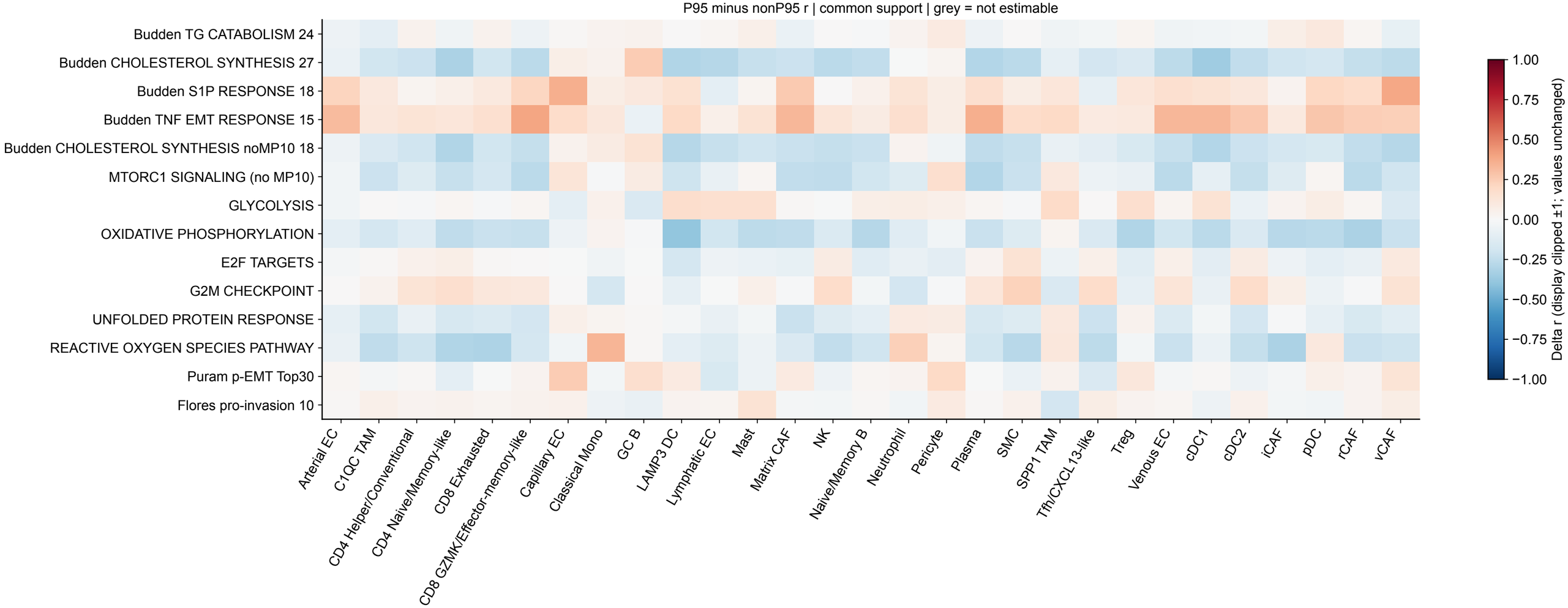
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

NPJ_sample5 | all 29 candidates; fixed core programs

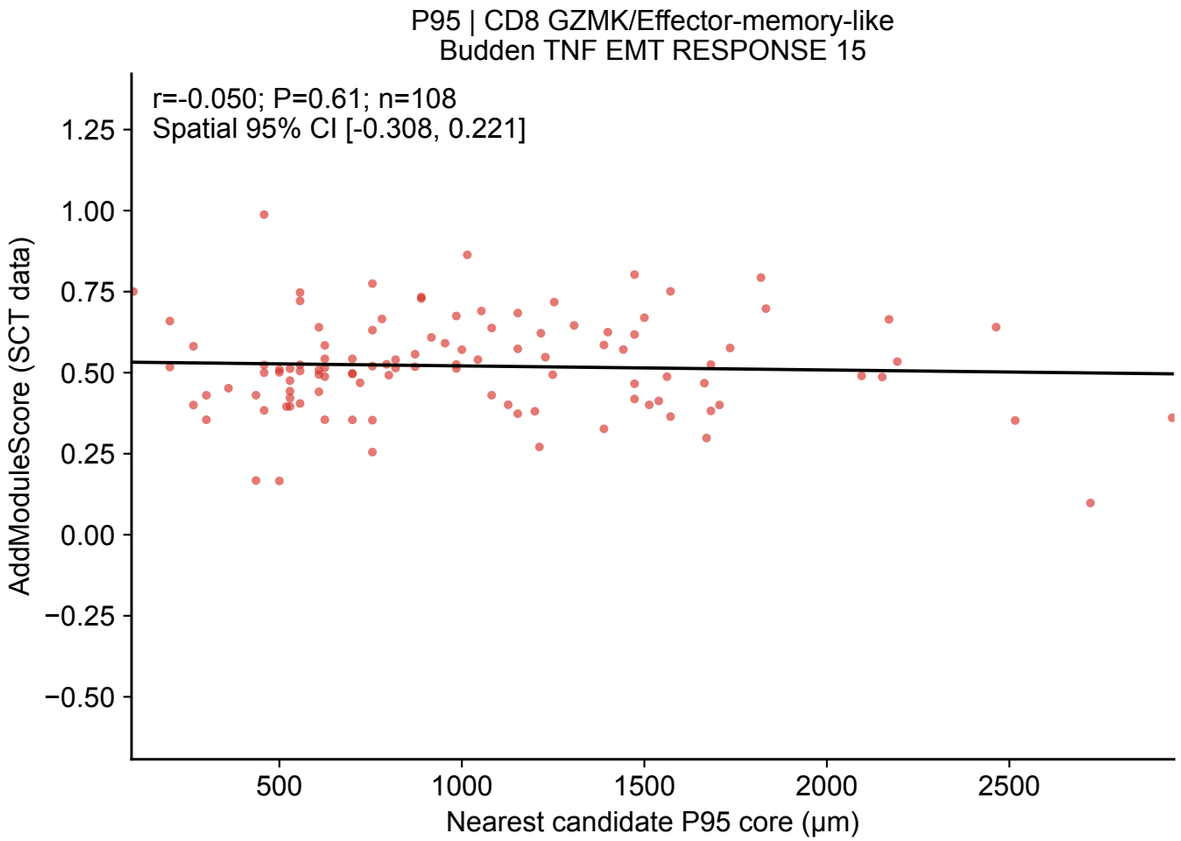
P95 raw Pearson r | full observed range

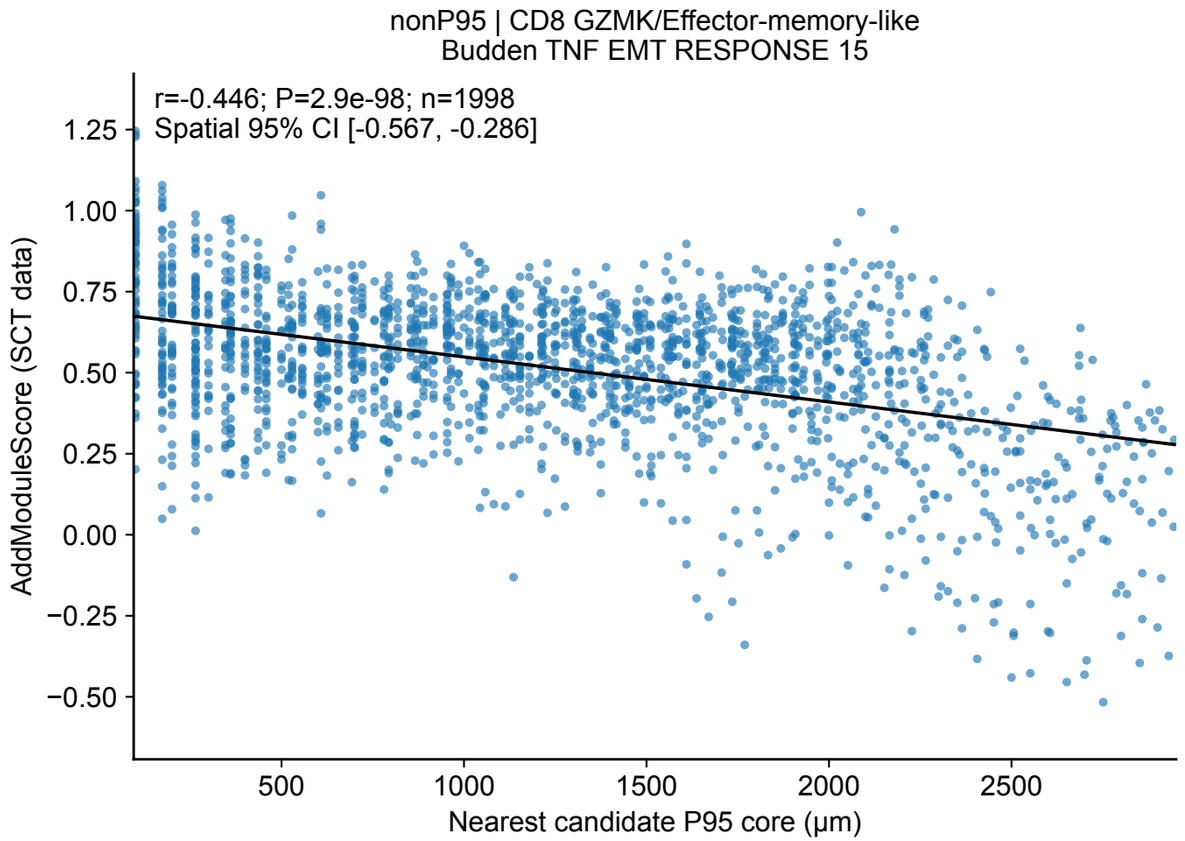


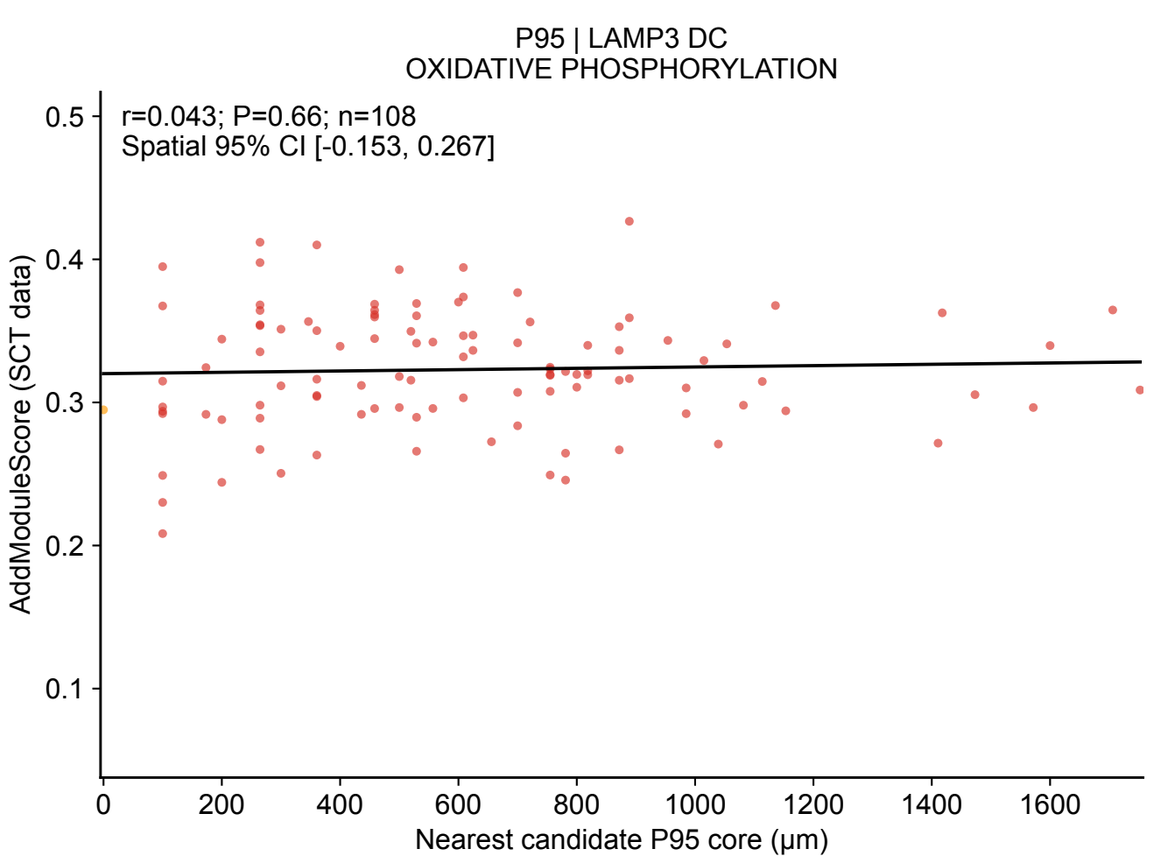
P95 minus nonP95 r | common support | grey = not estimable

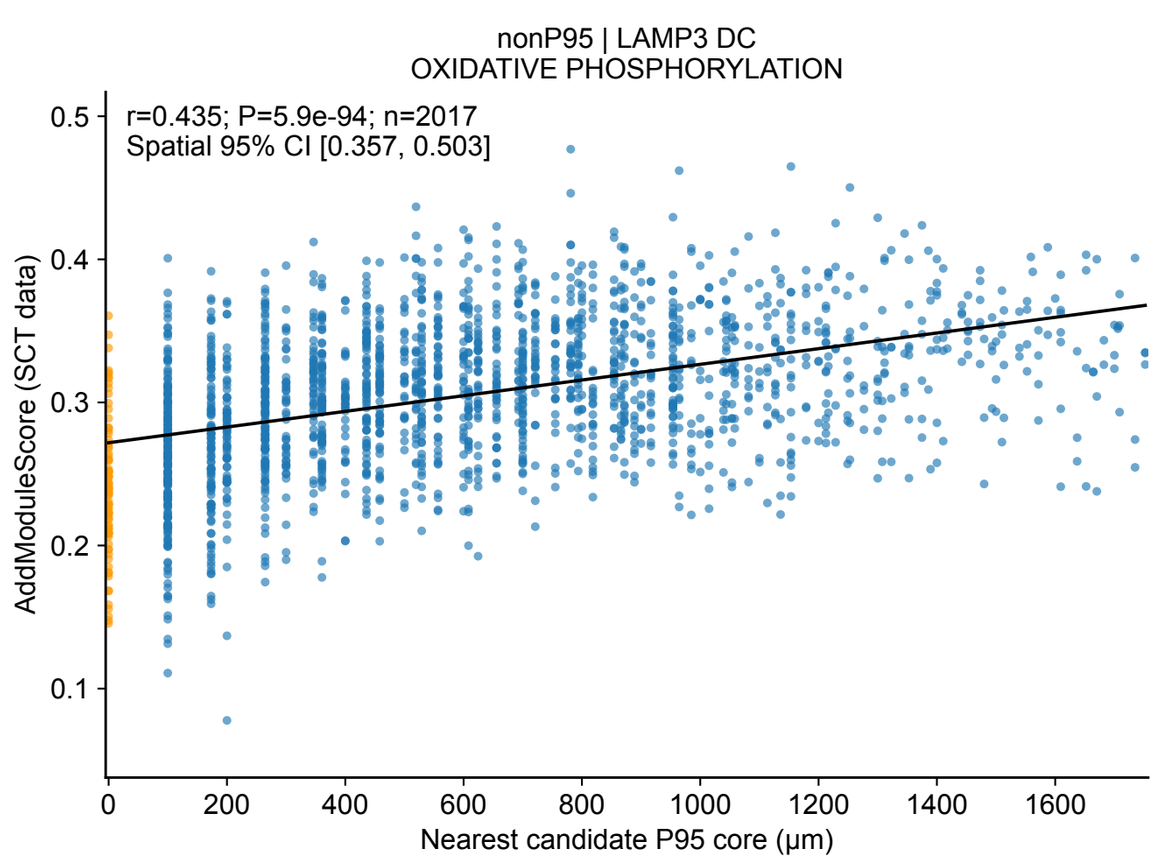


NPJ_sample5 | exploratory scatter display selected by coverage and |delta r|, not P





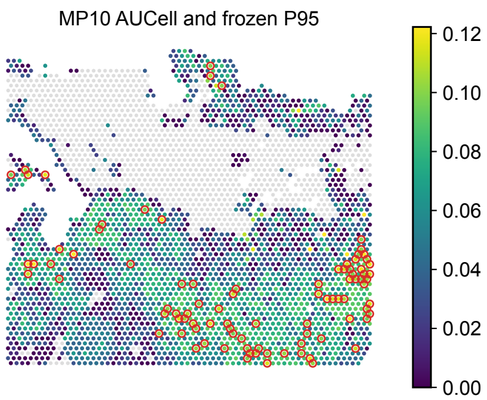


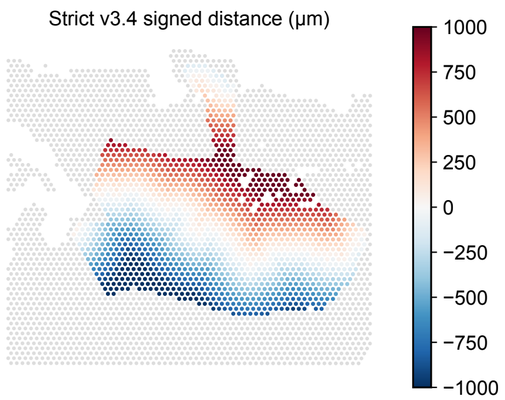


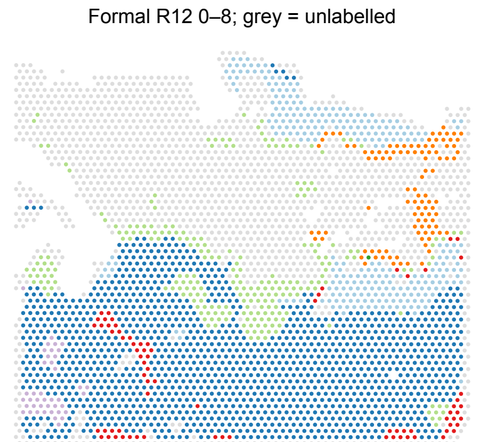
CD8 GZMK/Effector-memory-like / Budden TNF EMT RESPONSE 15: delta r=0.396 [0.162, 0.667]; common 100–2946 µm; spatial P=0.0013
LAMP3 DC / OXIDATIVE PHOSPHORYLATION: delta r=-0.393 [-0.582, -0.171]; common 0–1752 µm; spatial P=0.00018
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

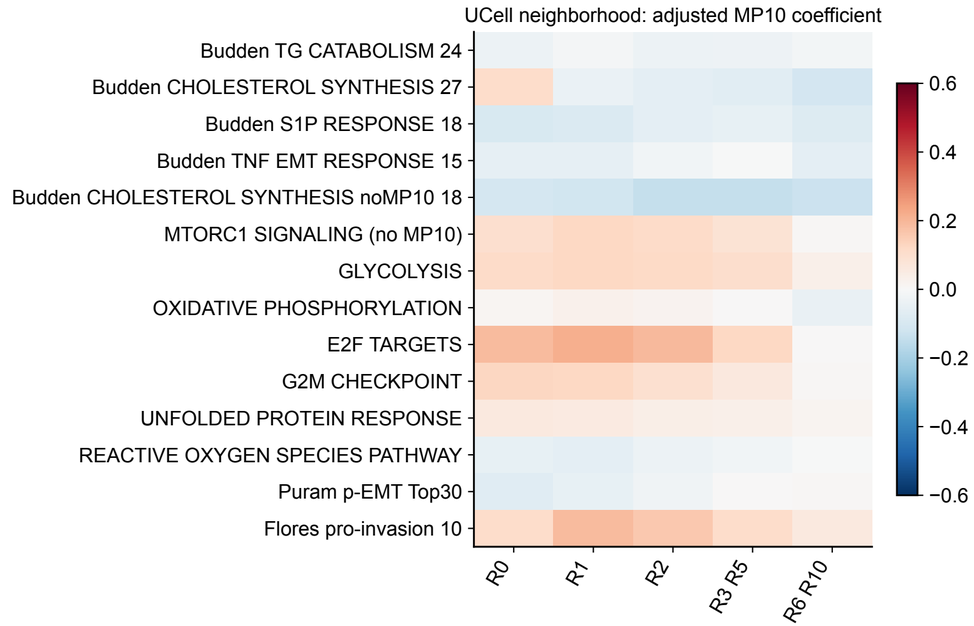
P95 nonP95 distance 0

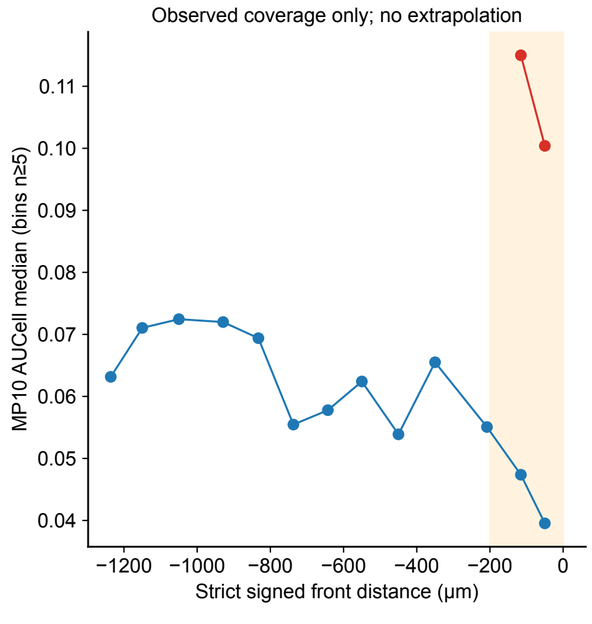
NPJ_sample6 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

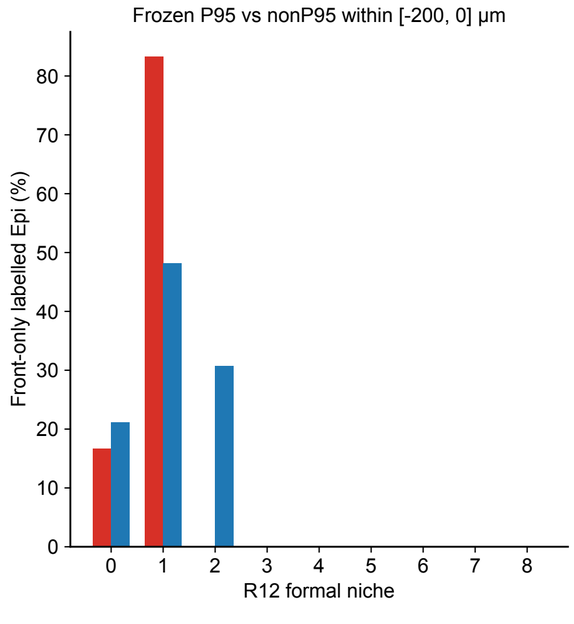








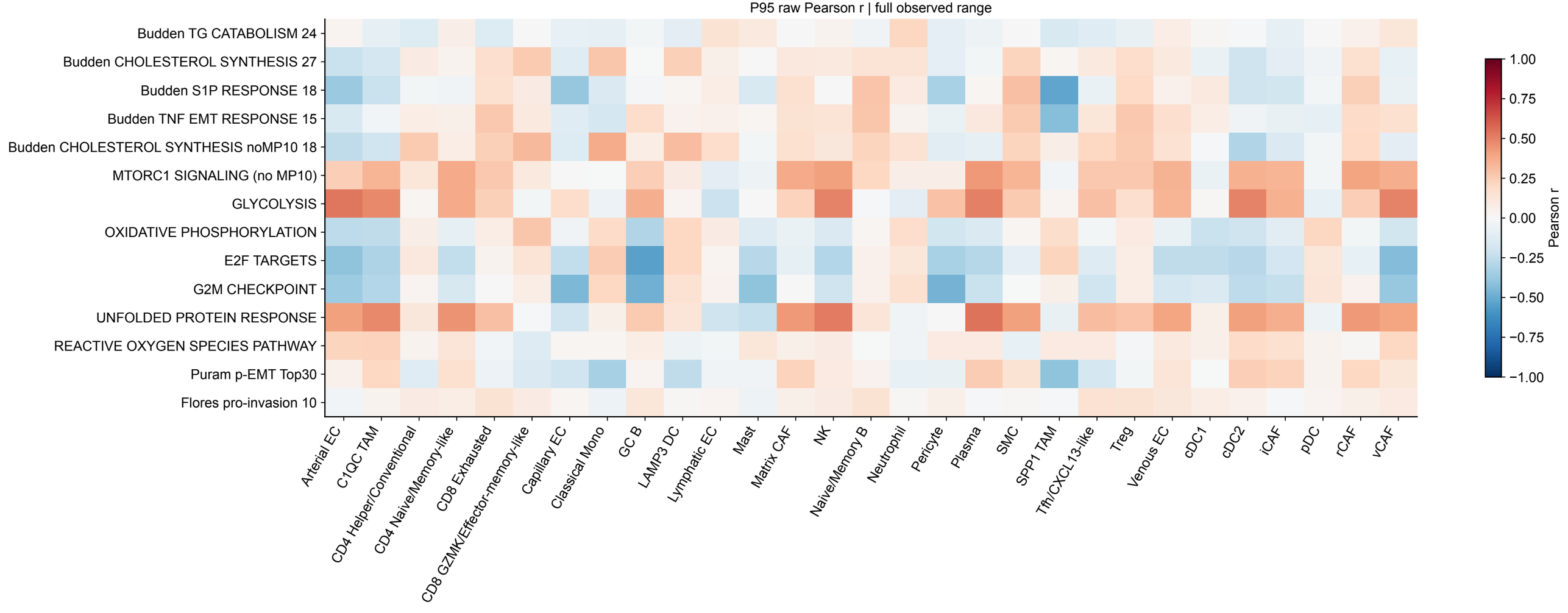




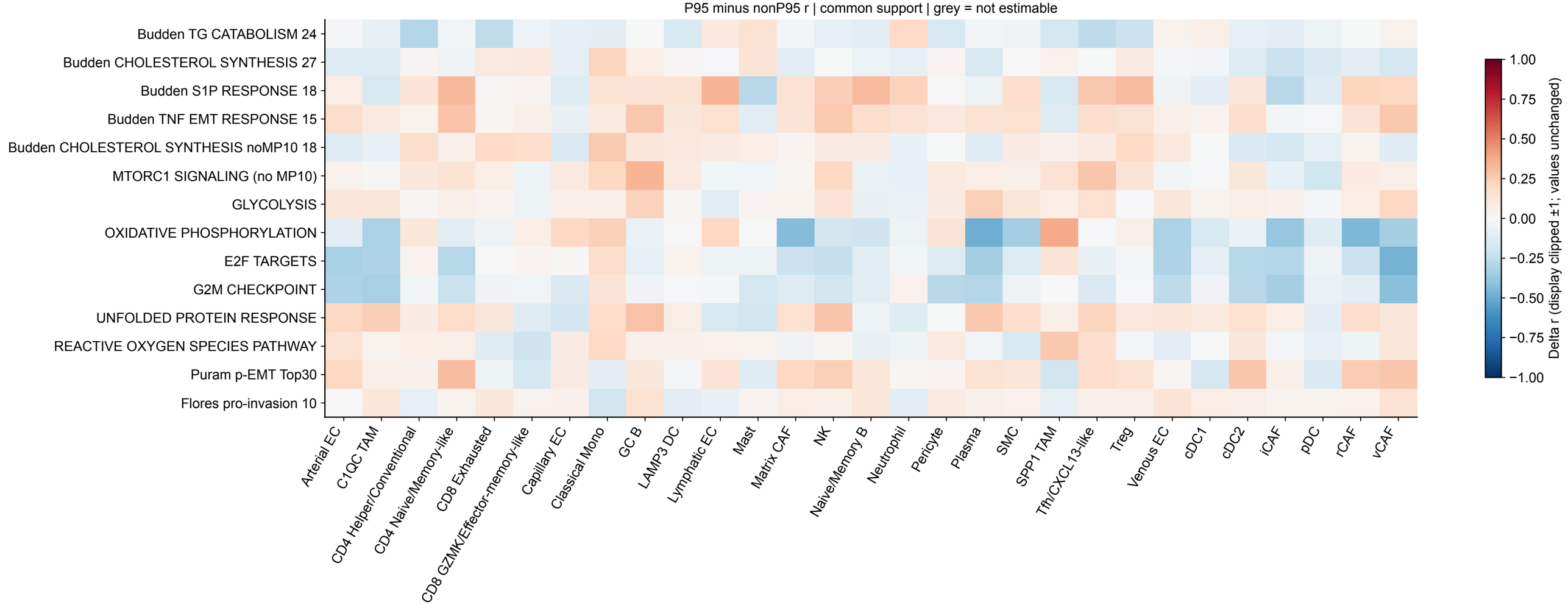
Tissue 3497 | Epi 2414 | internal 1808 | P95 91
Exact front in Epi 957 | R12 labelled 2072

NPJ_sample6 | all 29 candidates; fixed core programs

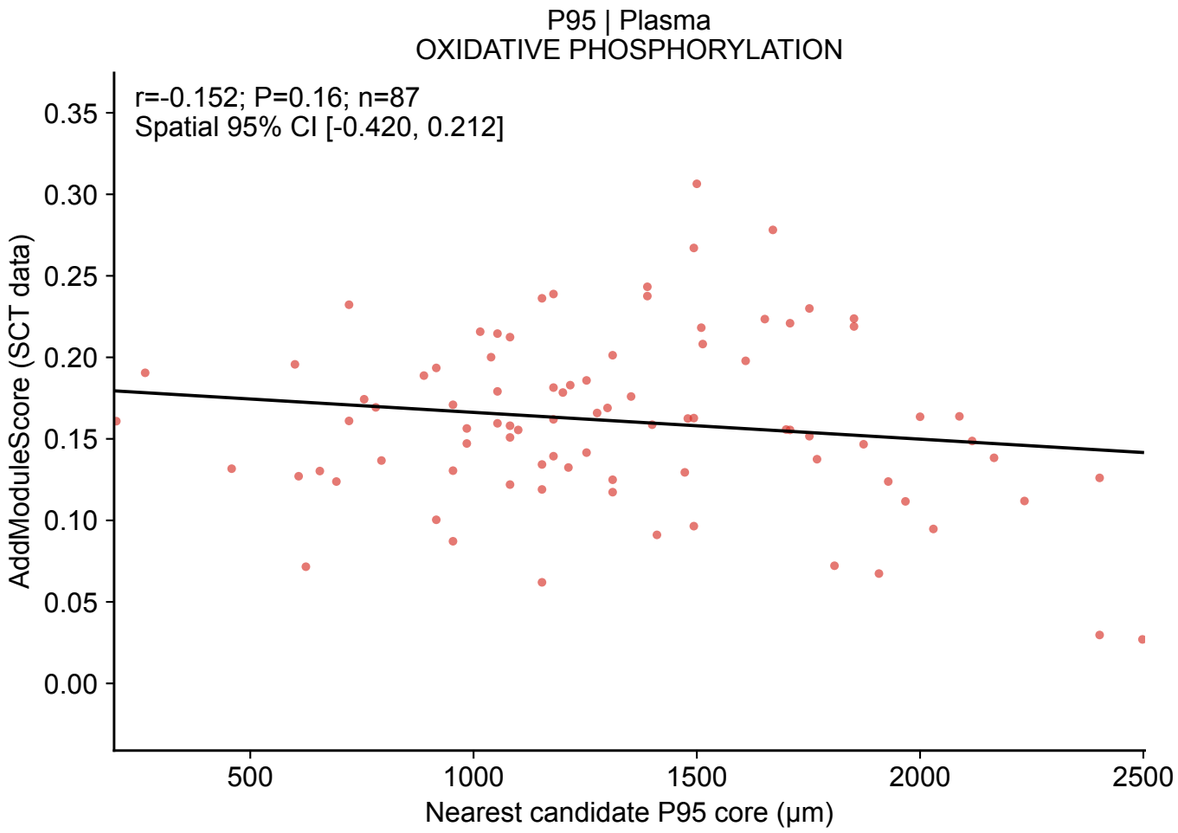
P95 raw Pearson r | full observed range

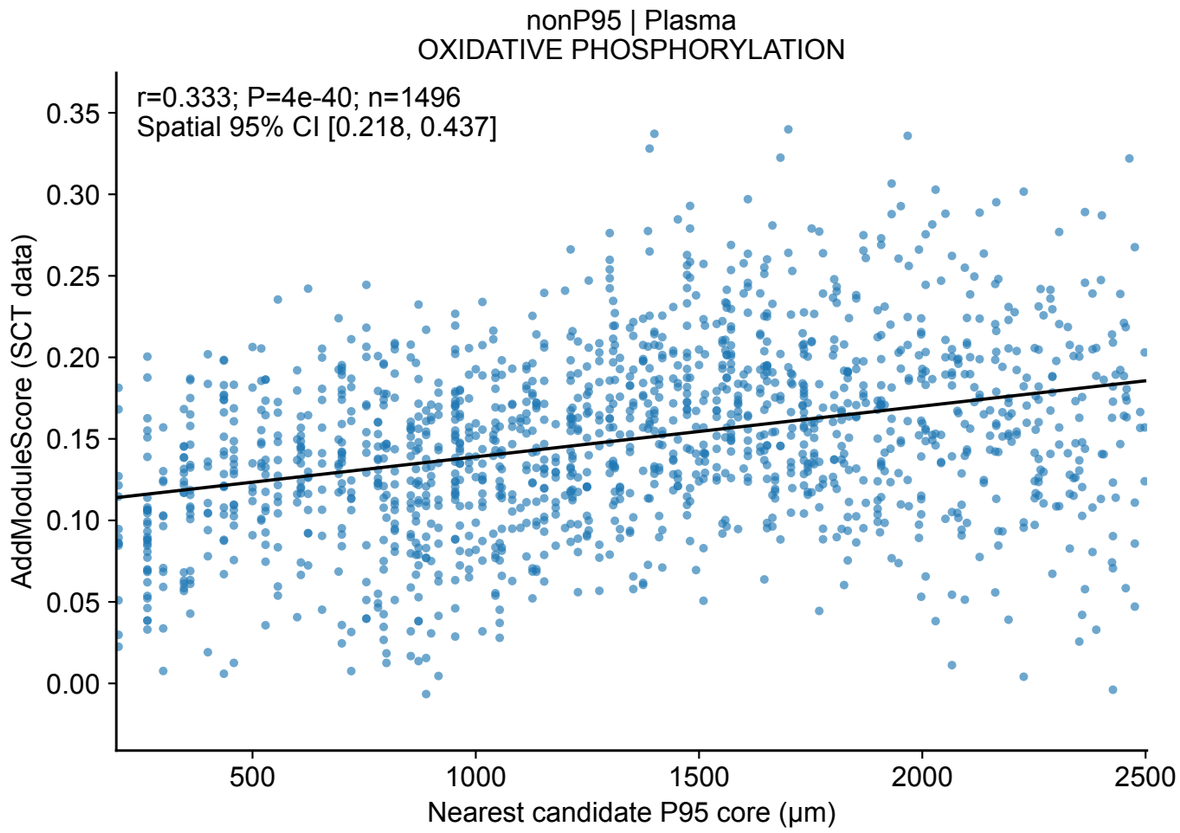


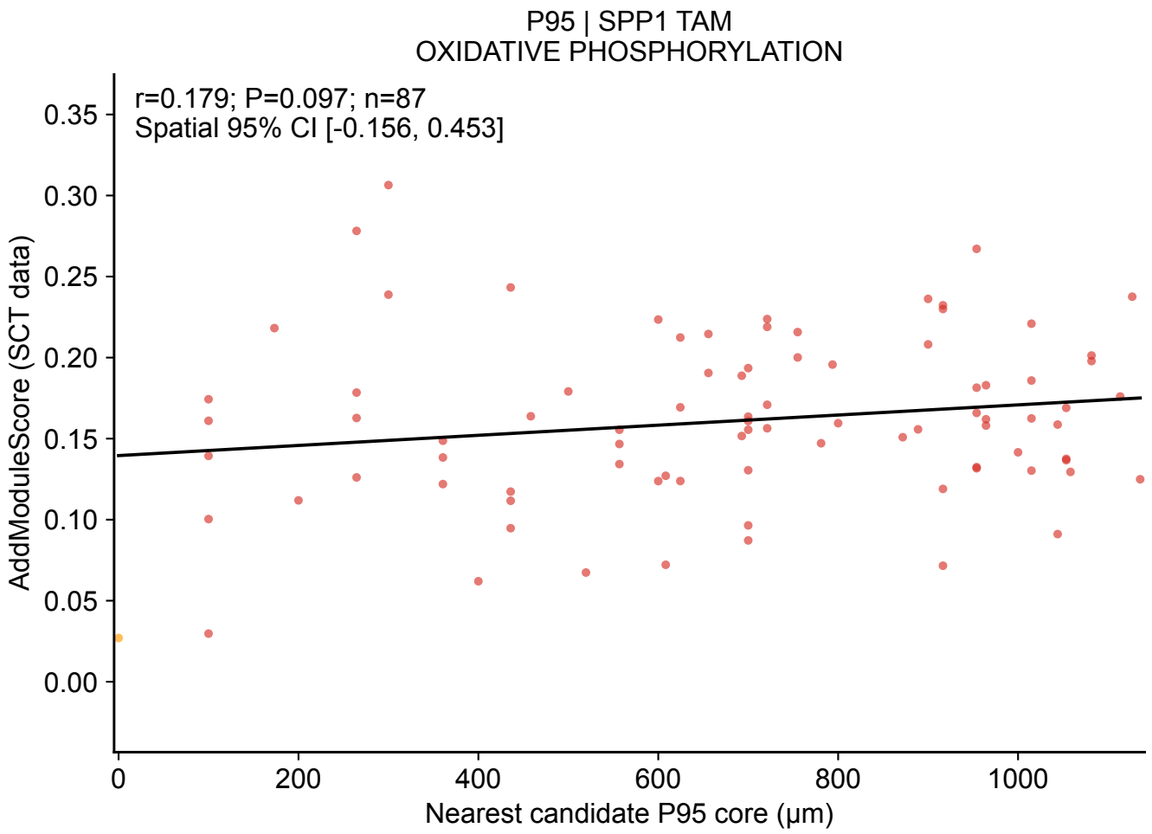
P95 minus nonP95 r | common support | grey = not estimable

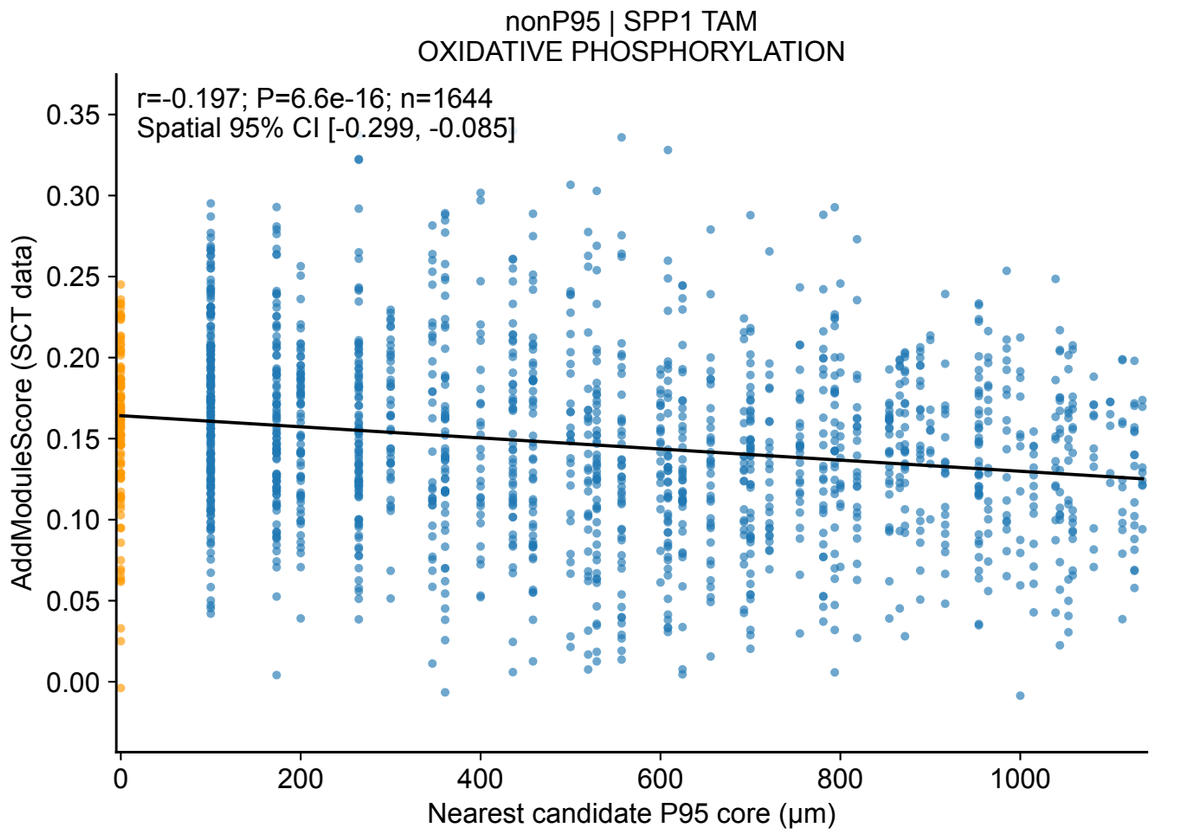


NPJ_sample6 | exploratory scatter display selected by coverage and |delta r|, not P





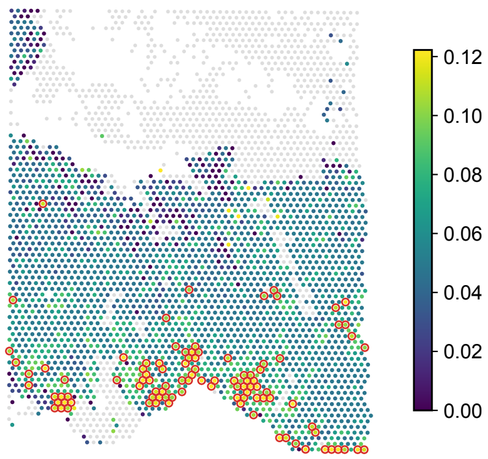


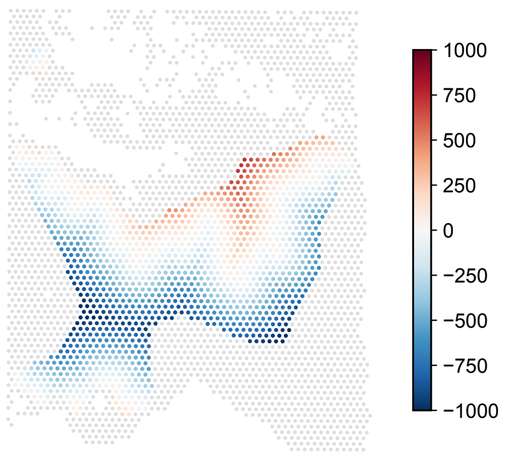


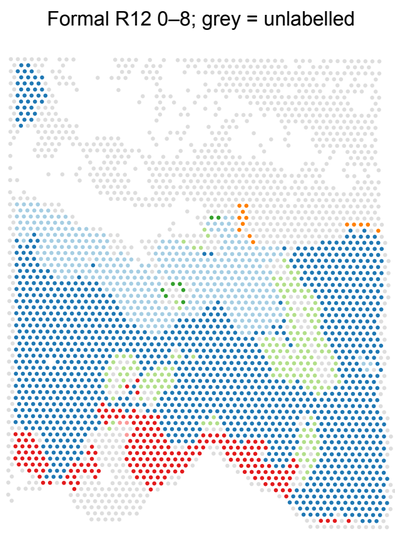
Plasma / OXIDATIVE PHOSPHORYLATION: delta r=-0.485 [-0.739, -0.151]; common 200–2498 µm; spatial P=0.0019
SPP1 TAM / OXIDATIVE PHOSPHORYLATION: delta r=0.377 [0.049, 0.641]; common 0–1136 µm; spatial P=0.02
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

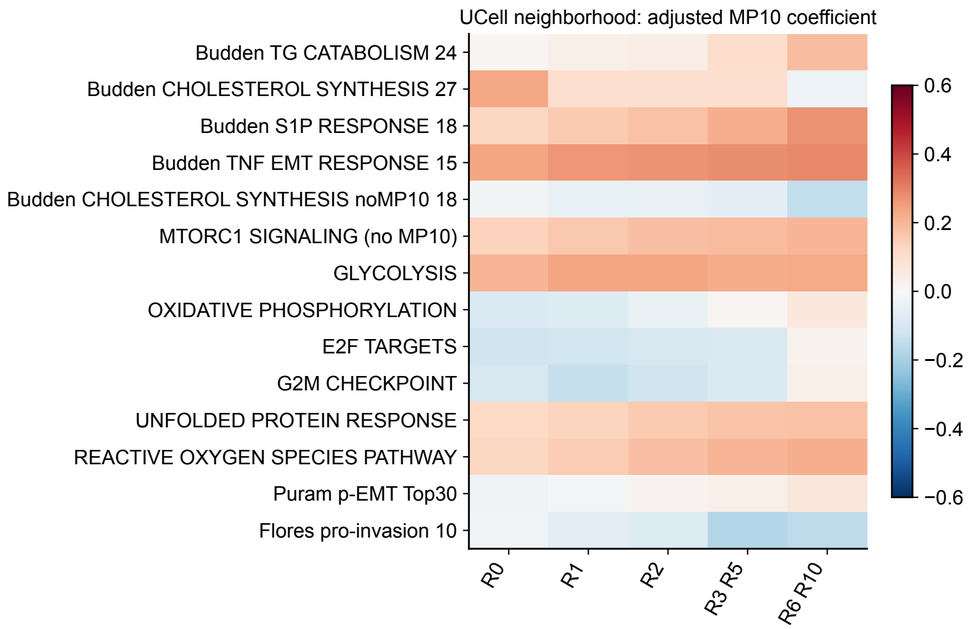
P95 nonP95 distance 0

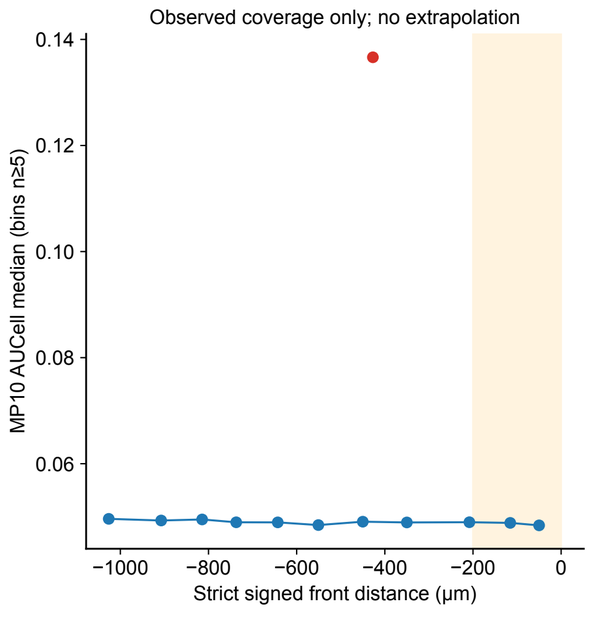
NPJ_sample7 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

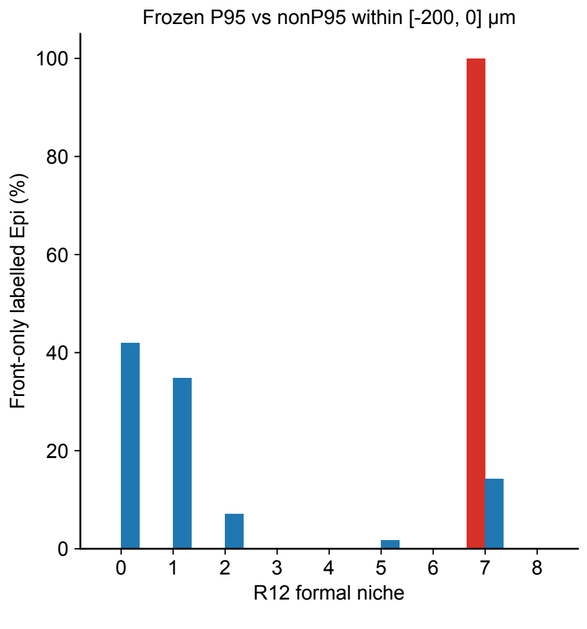










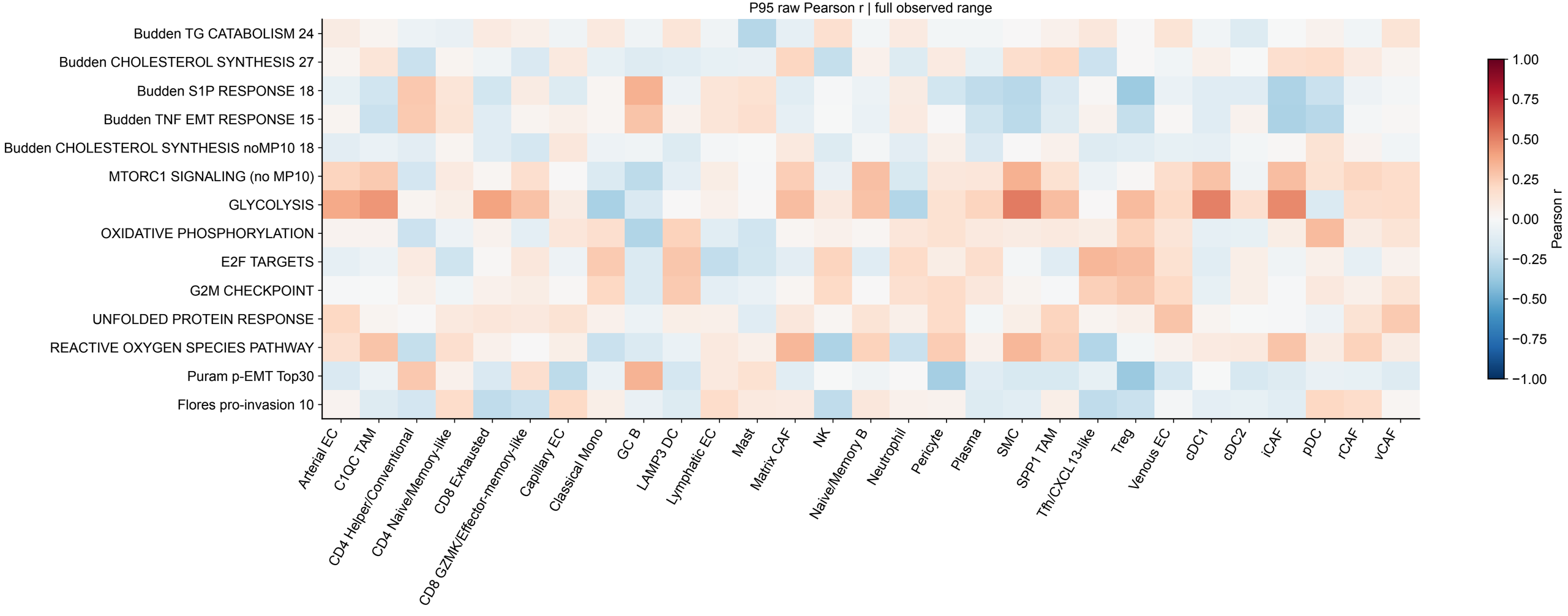


Tissue 3386 | Epi 2382 | internal 1957 | P95 99
Exact front in Epi 1178 | R12 labelled 2129

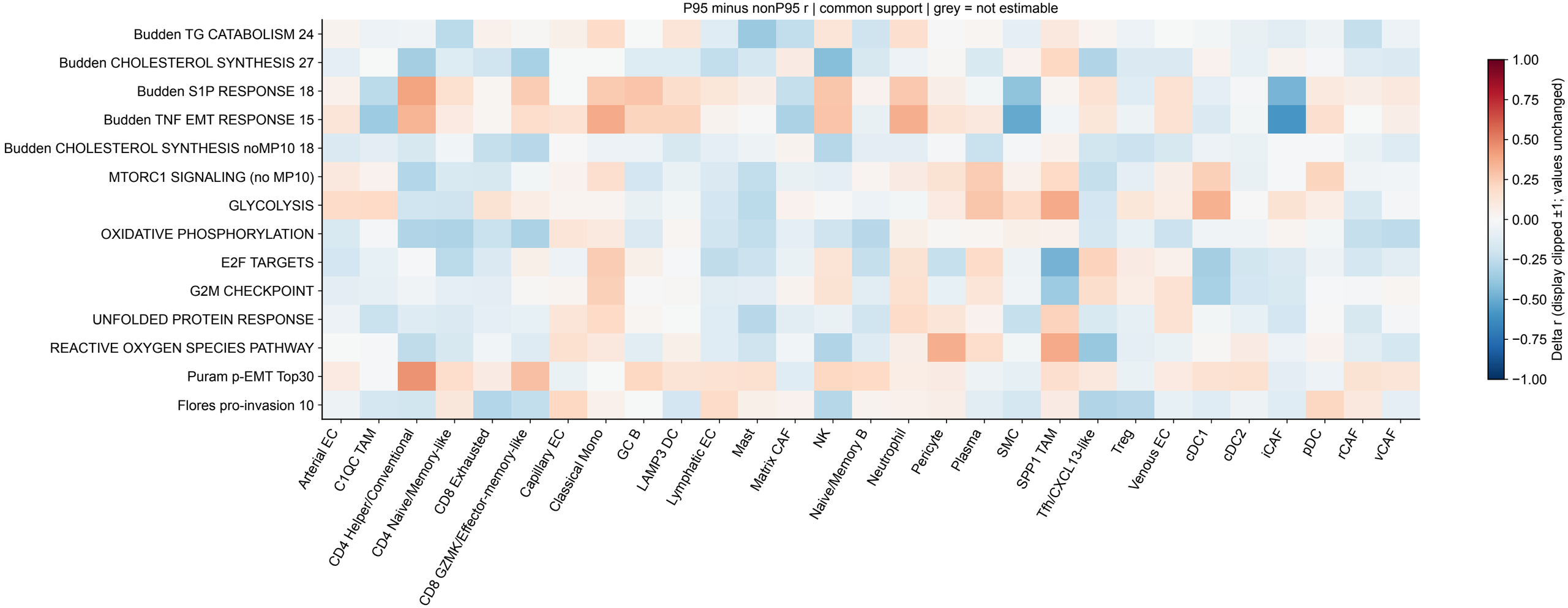
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

NPJ_sample7 | all 29 candidates; fixed core programs

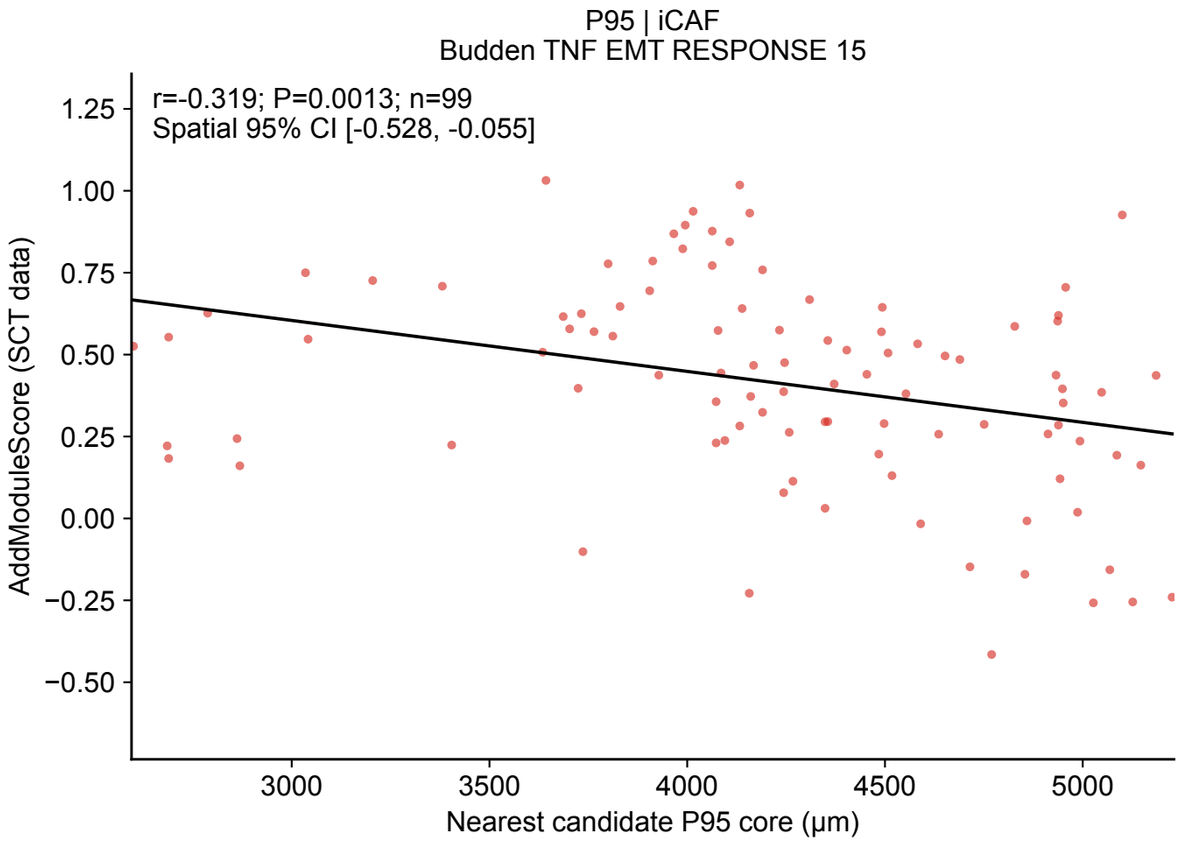
P95 raw Pearson r | full observed range

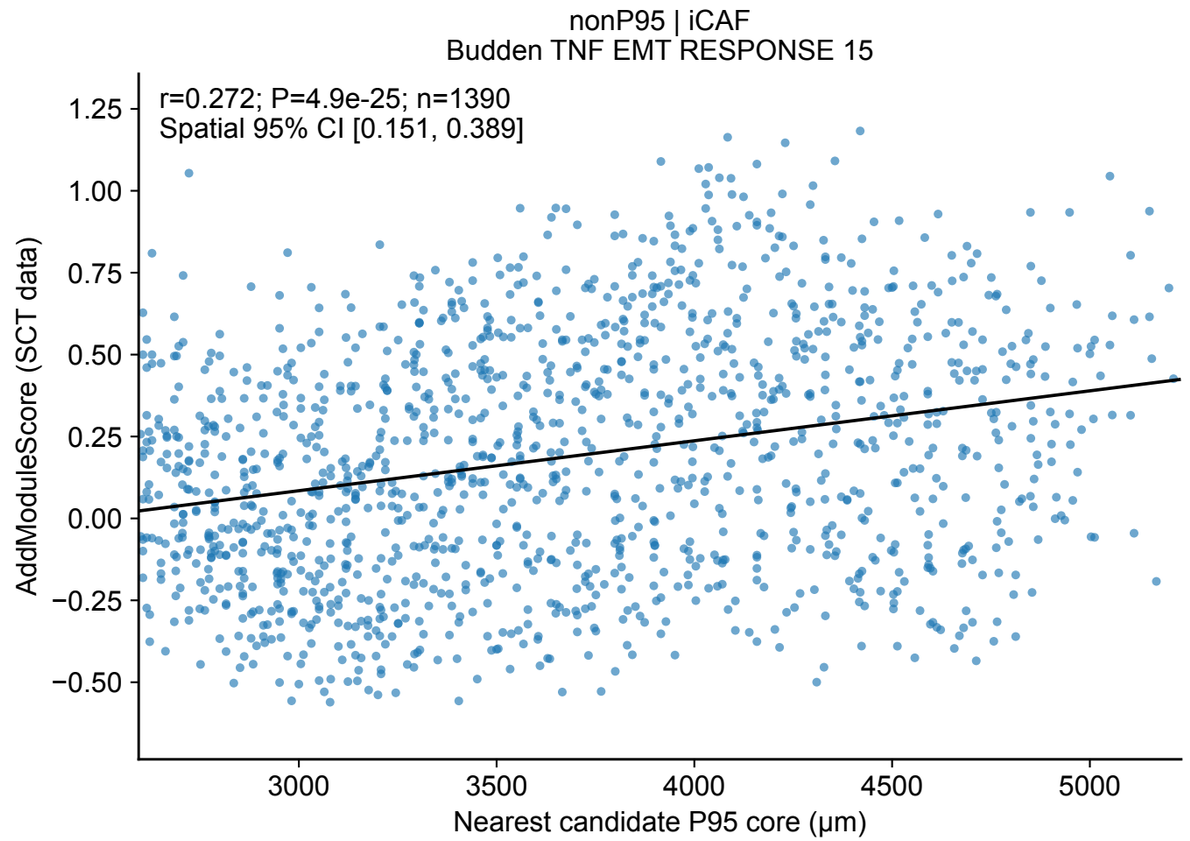


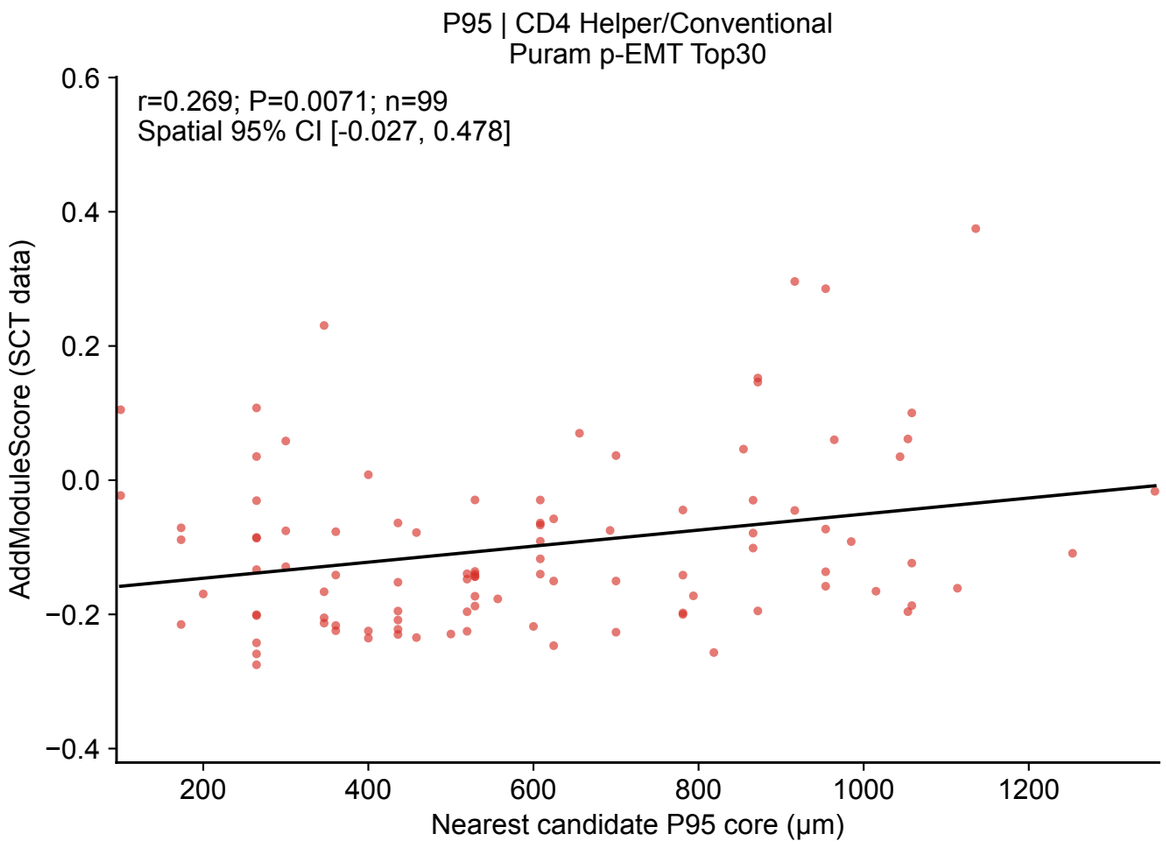
P95 minus nonP95 r | common support | grey = not estimable

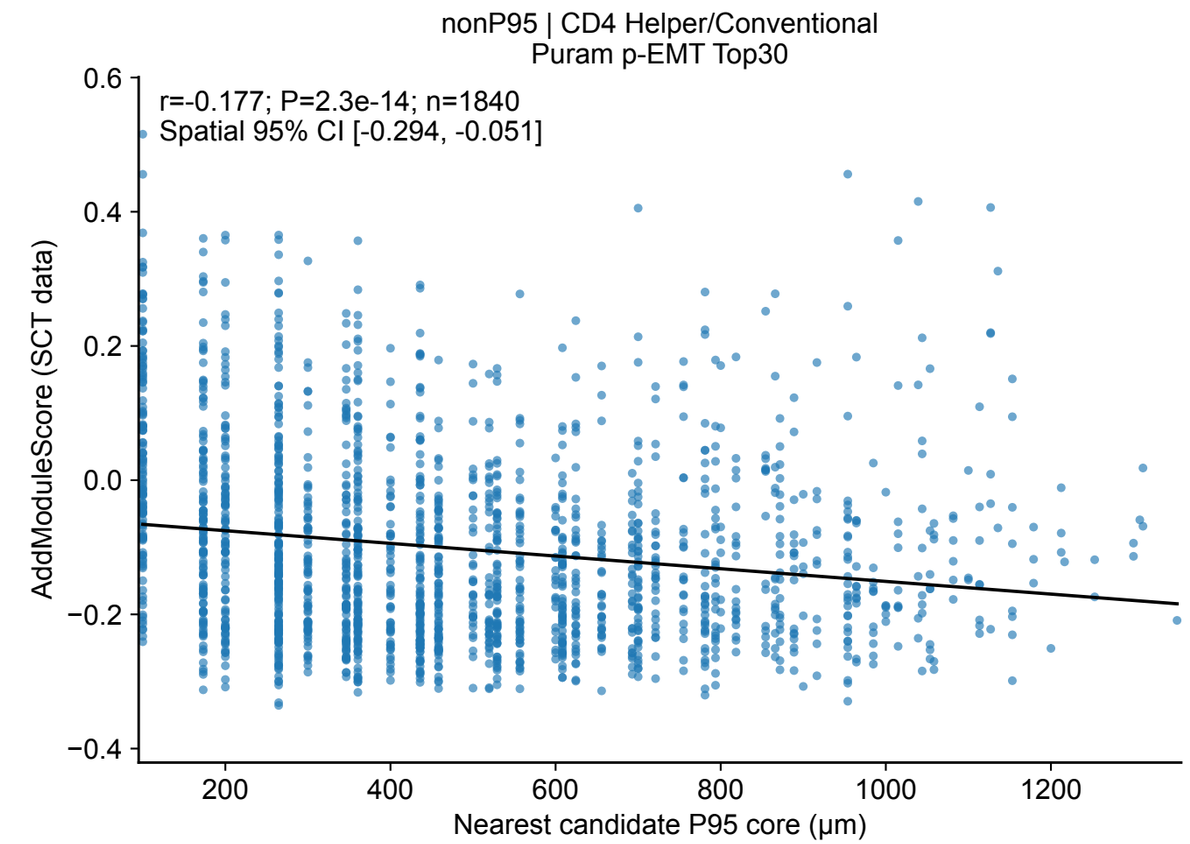


NPJ_sample7 | exploratory scatter display selected by coverage and |delta r|, not P



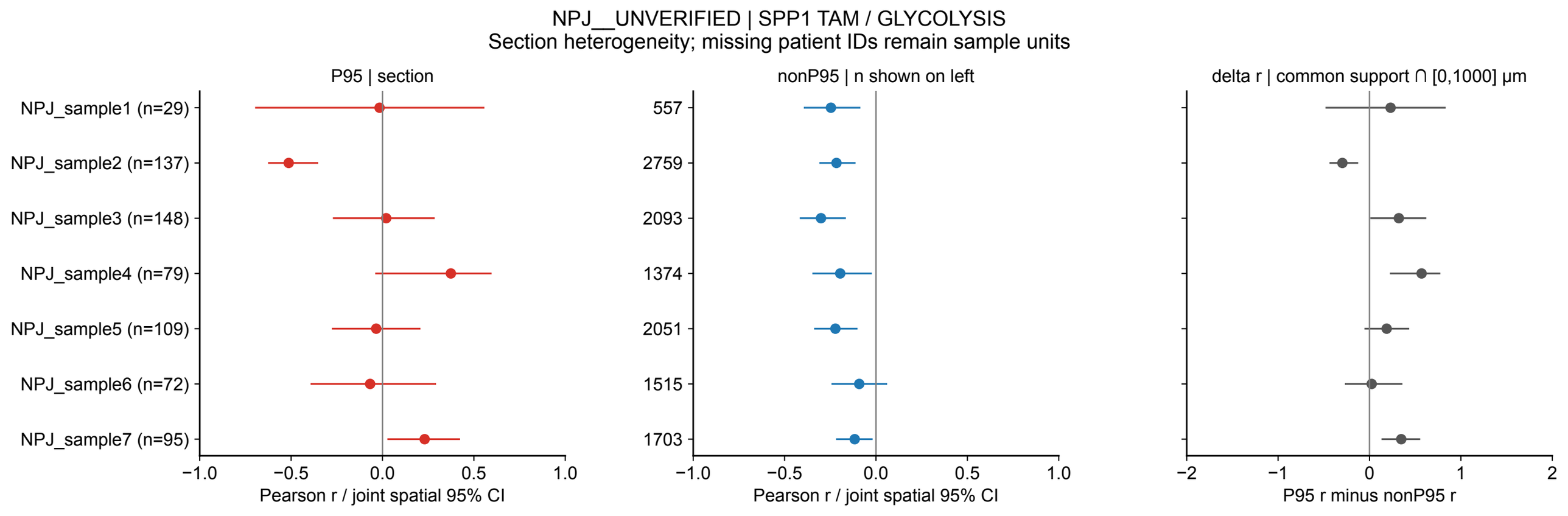


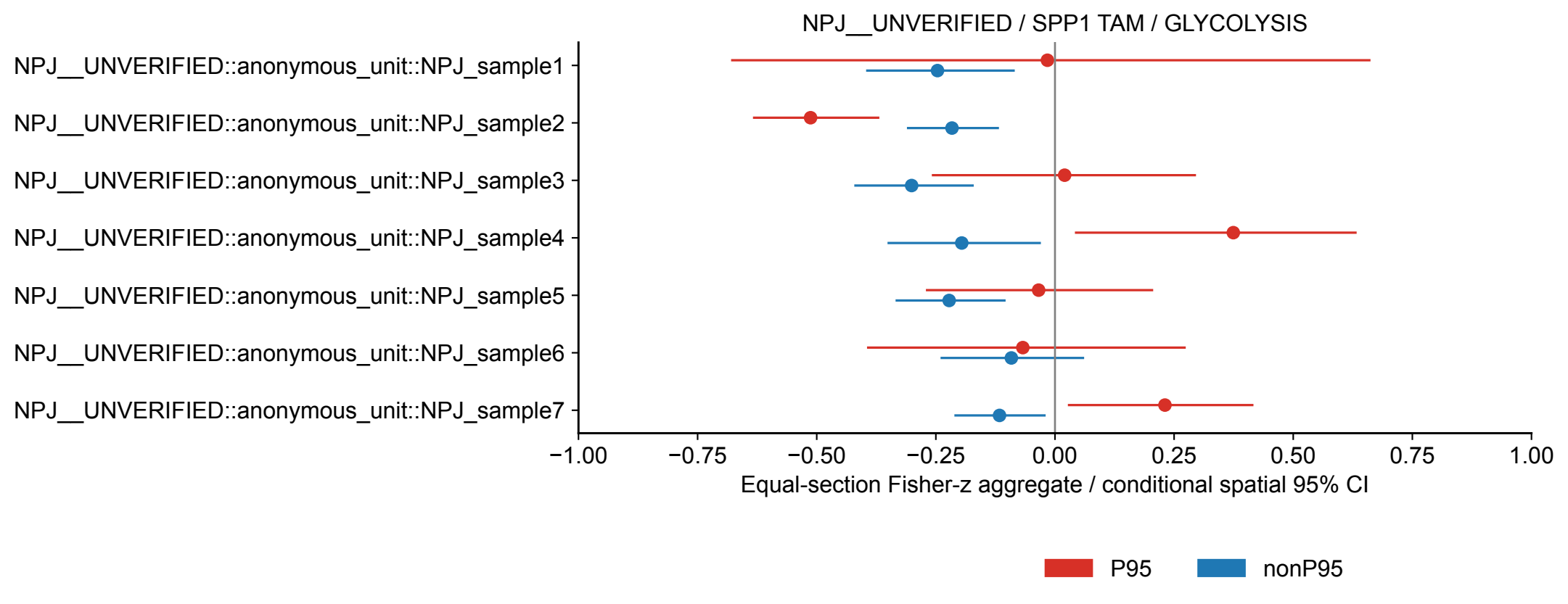


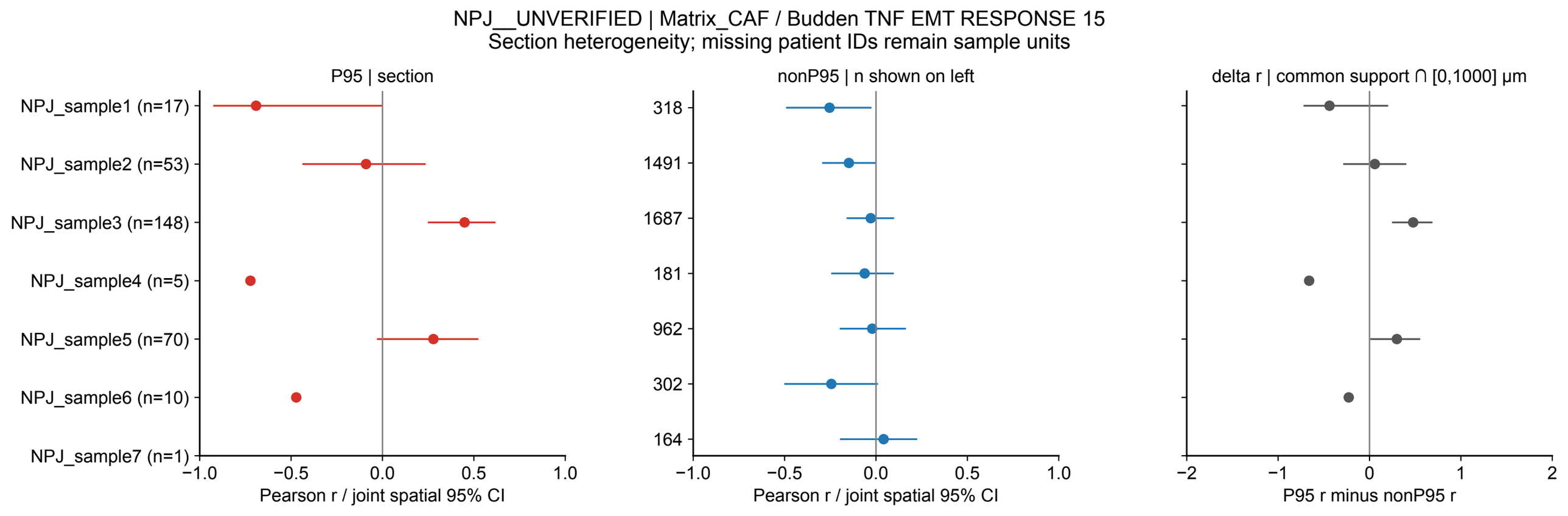


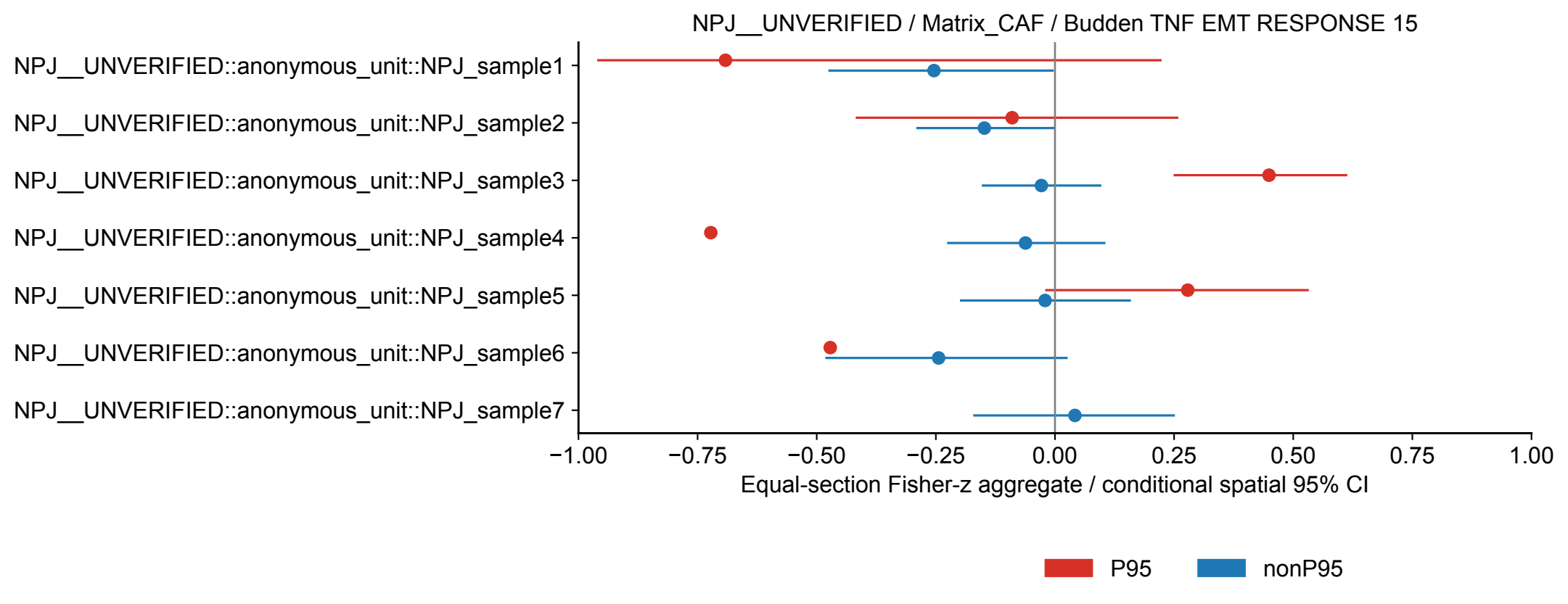
iCAF / Budden TNF EMT RESPONSE 15: delta r=-0.591 [-0.810, -0.312]; common 2600–5226 µm; spatial P=2.8e-05
CD4 Helper/Conventional / Puram p-EMT Top30: delta r=0.446 [0.185, 0.639]; common 100–1353 µm; spatial P=9.2e-05
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

P95 nonP95 distance 0


NPJ__UNVERIFIED | SPP1 TAM / GLYCOLYSIS
Section heterogeneity; missing patient IDs remain sample units
P95 | section
nonP95 | n shown on left
delta r | common support ∩ [0,1000] µm
NPJ_sample1 (n=29)
NPJ_sample2 (n=137)
NPJ_sample3 (n=148)
NPJ_sample4 (n=79)
NPJ_sample5 (n=109)
NPJ_sample6 (n=72)
NPJ_sample7 (n=95)
557
2759
2093
1374
2051
1515
1703
−1.0
−0.5
0.0
0.5
1.0
−2
−1
0
1
2
Pearson r / joint spatial 95% CI
Pearson r / joint spatial 95% CI
P95 r minus nonP95 r


NPJ__UNVERIFIED / SPP1 TAM / GLYCOLYSIS
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample1
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample2
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample3
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample4
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample5
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample6
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample7
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Equal-section Fisher-z aggregate / conditional spatial 95% CI
P95
nonP95


NPJ__UNVERIFIED | Matrix_CAF / Budden TNF EMT RESPONSE 15
Section heterogeneity; missing patient IDs remain sample units
P95 | section
NPJ_sample1 (n=17)
NPJ_sample2 (n=53)
NPJ_sample3 (n=148)
NPJ_sample4 (n=5)
NPJ_sample5 (n=70)
NPJ_sample6 (n=10)
NPJ_sample7 (n=1)
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
nonP95 | n shown on left
318
1491
1687
181
962
302
164
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
delta r | common support ∩ [0,1000] µm
−2
−1
0
1
2
P95 r minus nonP95 r


NPJ__UNVERIFIED / Matrix_CAF / Budden TNF EMT RESPONSE 15
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample1
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample2
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample3
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample4
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample5
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample6
NPJ__UNVERIFIED::anonymous_unit::NPJ_sample7
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Equal-section Fisher-z aggregate / conditional spatial 95% CI
P95
nonP95